Fisher, F. (1970) "Tests on Equality Between Sets of Coefficients in Two Linear Regressions: An Expository Note." *Econometrica*, 28, 361–366.

Giles, D. (1982) "The Interpretation of Dummy Variables in Semilogarithmic Equations." *Economics Letters*, 10, 77–79.

Goldfeld, S., and R. Quandt (1973) "The Estimation of Structural Shifts by Switching Regressions." *Annals of Economic and Social Measurement*, 2, 475–485.

Griffiths, W. E. (1972) "Estimation of Actual Response Coefficients in the Hildreth–Houck Random Coefficients Model." *Journal of the American Statistical Association*, 67, 633–635.

Gujarati, D. (1978) *Basic Econometrics.* New York: McGraw-Hill.

Halvorsen, R., and R. Palmquist (1980) "The Interpretation of Dummy Variables in Semilogarithmic Equations." *American Economic Review*, 70, 474–475.

Hildreth, C., and J. Houck (1968) "Some Estimators for a Linear Model with Random Coefficients." *Journal of the American Statistical Association*, 63, 584–595.

Houck, J. (1977) "An Approach to Specifying and Estimating Nonreversible Functions," *American Journal of Agricultural Economics*, 51, 570–572.

Johnston, J. (1984) *Econometric Methods*, 3rd ed. New York: McGraw-Hill.

Judge, G. G., W. E. Griffiths, R. C. Hill, H. Lütkepohl, T. C. Lee (1985) *The Theory and Practice of Econometrics*, 2nd ed. New York: Wiley.

Kennedy, P. (1981) "Estimation with Correctly Interpreted Dummy Variables in Semilogarithmic Equations." *American Economic Review*, 71, 801.

Kmenta, J. (1986) *Elements of Econometrics*, 2nd ed. New York: Macmillan.

Nicholls, D., and A. Pagan (1985) "Varying Coefficient Regression." In E. Hannan, P. Krishnaiah, and M. Rao, (eds.). *Handbook of Statistics*, Vol. 5, Chapter 16. Amsterdam: Elsevier.

Raj, B., and A. Ullah (1981) *Econometrics, A Varying Coefficients Approach.* London: Croom-Helm.

CHAPTER 11

Sets of Linear Statistical Models

11.1 Introduction

In most of the earlier chapters we concentrated on estimation and hypothesis testing for the parameter vector $\boldsymbol{\beta}$ in the single equation model $\mathbf{y} = X\boldsymbol{\beta} + \mathbf{e}$. Different estimation methods and tests were considered, depending on what further assumptions were made about the disturbance covariance matrix $E[\mathbf{ee}'] = \sigma^2\boldsymbol{\Psi}$. In this Chapter we turn to a situation where there is more than one equation to estimate. For example, interest might center on demand equations for a number of commodities, investment functions for a number of firms, or consumption functions for subsets of the population. Suppose that time-series data suitable for estimation of a number of demand equations are available. The disturbances in these different equations at a given time are likely to reflect some common unmeasurable or omitted factors, and hence could be correlated. The same is also likely to be true when time-series observations are used to estimate different investment functions for different firms, different consumption functions for different subsets of the population, or many other examples. With investment functions for a number of firms, the general state of the economy (a variable often not explicitly included) is likely to have similar effects on the disturbances of the different functions. In the study of agricultural supply response, the effect of weather in a given year is likely to have related effects on the disturbances for different crops. Correlation between disturbances from different equations at a given time is known as *contemporaneous correlation.* It is distinct from "autocorrelation," which, as we saw in a previous chapter, refers to correlation over time for the disturbances in a single equation. When contemporaneous correlation exists, it may be more efficient to estimate all equations jointly, rather than to estimate each one separately using least squares. The appropriate joint estimation technique is often known as seemingly unrelated regressions estimation; it is considered in detail in Section 11.2. Although seemingly unrelated regressions are usually described in the context of estimating a number of equations using time-series data, they can be equally relevant for cross-sectional data. If cross-sectional household data were used to estimate expenditure functions for a number of commodities, then it is quite likely that some unmeasurable characteristics of a given household could have similar effects on the disturbances of all the expenditure functions.

In many instances the use of seemingly unrelated regressions can be regarded as one method for pooling time-series and cross-sectional data. For example, if time-series data on a number of firms are used to estimate an investment function for each firm, we are, in effect, pooling the cross-section of firms with the time-series data on each firm. The distinguishing features of seemingly unrelated regressions as a method for pooling time-series and cross-sectional data are contemporaneous correlation in the disturbances and the assumption that each cross-sectional unit has a different coefficient vector. Other pooling methods make alternative assumptions about the disturbance terms and about how the coefficients change over cross-sections or time. One method that is more restrictive than seemingly unrelated regressions because it does not entertain the possibility of contemporaneous correlation, and because it assumes all cross-sectional units have identical coefficient vectors except for the intercept term, is the dummy variable or covariance model. In this model differences over cross-sectional units are assumed to be reflected in the intercept term; the cross-sectional units provide natural partitions for which dummy variables can be defined. This model is discussed in Section 11.4. A third pooling method, known as the error components model, is outlined in Section 11.5. In this model all cross-sectional units are assumed to have the same coefficient vector, but the disturbance term is a composite one made up of a unit-specific random element and another element that is random over both time and cross-sections. The question of choosing between alternative models is addressed in Section 11.6.

11.2 Seemingly Unrelated Regression Equations

As a vehicle for introducing contemporaneous correlation and the conditions under which it is possible to improve on separate least squares estimation of a number of equations, let us consider the following set of three log-linear demand equations:

$$\begin{aligned}
\ln q_{1t} &= \beta_{10} + \beta_{11} \ln p_{1t} + \beta_{14} \ln y_t + e_{1t} \\
\ln q_{2t} &= \beta_{20} + \beta_{22} \ln p_{2t} + \beta_{24} \ln y_t + e_{2t} \\
\ln q_{3t} &= \beta_{30} + \beta_{33} \ln p_{3t} + \beta_{34} \ln y_t + e_{3t}
\end{aligned} \tag{11.2.1}$$

It is assumed that quantity demanded of the ith commodity q_i depends on own price p_i and income y. It would be more realistic to include other prices in each equation; and, indeed, in Exercise 11.3.7 we include p_1, p_2, and p_3 in all equations. For the moment, the specification in (11.2.1) is more convenient. The subscripts on the β's reflect the fact that $\ln p_2$ and $\ln p_3$ have been omitted from the first equation ($\beta_{12} = \beta_{13} = 0$), $\ln p_1$ and $\ln p_3$ have been omitted from the second equation ($\beta_{21} = \beta_{23} = 0$), and $\ln p_1$ and $\ln p_2$ have been omitted from the third equation ($\beta_{31} = \beta_{32} = 0$).

The three demand equations can be written in the usual matrix algebra notation as

$$\begin{aligned}
\mathbf{y}_1 &= X_1\boldsymbol{\beta}_1 + \mathbf{e}_1 \\
\mathbf{y}_2 &= X_2\boldsymbol{\beta}_2 + \mathbf{e}_2 \\
\mathbf{y}_3 &= X_3\boldsymbol{\beta}_3 + \mathbf{e}_3
\end{aligned} \tag{11.2.2}$$

where

$$\mathbf{y}_i = \begin{bmatrix} \ln q_{i1} \\ \ln q_{i2} \\ \vdots \\ \ln q_{iT} \end{bmatrix} \qquad X_i = \begin{bmatrix} 1 & \ln p_{i1} & \ln y_1 \\ 1 & \ln p_{i2} & \ln y_2 \\ \vdots & \vdots & \vdots \\ 1 & \ln p_{iT} & \ln y_T \end{bmatrix} \qquad i = 1, 2, 3 \tag{11.2.3}$$

$$\boldsymbol{\beta}_1 = \begin{bmatrix} \beta_{10} \\ \beta_{11} \\ \beta_{14} \end{bmatrix} \qquad \boldsymbol{\beta}_2 = \begin{bmatrix} \beta_{20} \\ \beta_{22} \\ \beta_{24} \end{bmatrix} \qquad \boldsymbol{\beta}_3 = \begin{bmatrix} \beta_{30} \\ \beta_{33} \\ \beta_{34} \end{bmatrix} \tag{11.2.4}$$

and

$$\mathbf{e}_i = \begin{bmatrix} e_{i1} \\ e_{i2} \\ \vdots \\ e_{iT} \end{bmatrix} \qquad i = 1, 2, 3 \tag{11.2.5}$$

Thus, $\mathbf{y}_1$ and X_1 contain all T observations on the dependent and explanatory variables in the demand equation for commodity 1, $\mathbf{y}_2$ and X_2 contain all T observations on the dependent and explanatory variables in the demand equation for commodity 2, and $\mathbf{y}_3$ and X_3 contain all T observations on the dependent and explanatory variables in the demand equation for commodity 3. Similarly, $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, and $\boldsymbol{\beta}_3$ are the (3×1) coefficient vectors for each of the equations and $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are the $(T \times 1)$ disturbance vectors for each of the equations.

The assumptions we will employ are

1. All disturbances have a zero mean,

$$E[e_{it}] = 0 \qquad i = 1, 2, 3; \quad t = 1, 2, \ldots, T \tag{11.2.6}$$

2. In a given equation the disturbance variance is constant over time, but each equation can have a different variance,

$$\left.\begin{aligned} \operatorname{var}(e_{1t}) &= E[e_{1t}^2] = \sigma_1^2 = \sigma_{11} \\ \operatorname{var}(e_{2t}) &= E[e_{2t}^2] = \sigma_2^2 = \sigma_{22} \\ \operatorname{var}(e_{3t}) &= E[e_{3t}^2] = \sigma_3^2 = \sigma_{33} \end{aligned}\right\} \quad t = 1, 2, \ldots, T \qquad (11.2.7)$$

The notation σ_{ii} is just an alternative way of writing σ_i^2.

3. Two disturbances in different equations but corresponding to the same time period are correlated (contemporaneous correlation),

$$\operatorname{covar}(e_{it}e_{jt}) = E[e_{it}e_{jt}] = \sigma_{ij} \qquad i, j = 1, 2, 3 \qquad (11.2.8)$$

4. Disturbances in different time periods, whether they are in the same equation or not, are uncorrelated (autocorrelation does not exist),

$$\operatorname{covar}(e_{it}e_{js}) = E[e_{it}e_{js}] = 0 \qquad \text{for } t \neq s \text{ and } i, j = 1, 2, 3 \quad (11.2.9)$$

In matrix notation these assumptions can be written compactly as

$$E[\mathbf{e}_i] = \mathbf{0} \quad \text{and} \quad E[\mathbf{e}_i\mathbf{e}_j'] = \sigma_{ij}I \qquad i, j = 1, 2, 3 \qquad (11.2.10)$$

Because $E[\mathbf{e}_1\mathbf{e}_1'] = \sigma_{11}I$, least squares applied to the first equation is the best linear unbiased estimator in the sense that it is the best estimator that is a linear unbiased function of $\mathbf{y}_1$. However, because of the existence of contemporaneous correlation, it is possible to obtain a better linear unbiased estimator that is a function of $\mathbf{y}_1$, $\mathbf{y}_2$, and $\mathbf{y}_3$. A similar result holds for the second and third equations. To illustrate how the improvement occurs, we write the three equations in (11.2.2) as the following "super model"

$$\underset{(3T \times 1)}{\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \end{bmatrix}} = \underset{(3T \times 9)}{\begin{bmatrix} X_1 & 0 & 0 \\ 0 & X_2 & 0 \\ 0 & 0 & X_3 \end{bmatrix}} \underset{(9 \times 1)}{\begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \boldsymbol{\beta}_3 \end{bmatrix}} + \underset{(3T \times 1)}{\begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \mathbf{e}_3 \end{bmatrix}} \qquad (11.2.11)$$

The matrix dimensions appear underneath each matrix; in the $(3T \times 9)$ matrix, "0" refers to a $(T \times 3)$ matrix of zeros.

Using obvious notation we can rewrite (11.2.11) as

$$\mathbf{y} = X\boldsymbol{\beta} + \mathbf{e} \qquad (11.2.12)$$

Thus we can write the three-equation "supermodel" in the framework of a single equation linear model with special definitions for $\mathbf{y}$, X, $\boldsymbol{\beta}$, and $\mathbf{e}$.

The natural question that now arises is whether $\boldsymbol{\beta}$, the vector containing the coefficients from all three equations, can be satisfactorily estimated by applying least squares, or some other technique, to (11.2.12). The first point to note is that it can be shown that least squares applied to the system in (11.2.12) is identical to applying least squares separately to each of the three equations (see Exercise 11.3.1). Thus, there is no benefit to be gained from applying least squares to the whole system.

To investigate whether generalized least squares is worthwhile, we consider the covariance matrix of the joint disturbance vector $\mathbf{e}$. This matrix is given by

$$\begin{aligned} \boldsymbol{\Phi} = E[\mathbf{e}\mathbf{e}'] &= E\left[\begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \mathbf{e}_3 \end{pmatrix} (\mathbf{e}_1' \quad \mathbf{e}_2' \quad \mathbf{e}_3')\right] \\ &= \begin{bmatrix} E\mathbf{e}_1\mathbf{e}_1' & E\mathbf{e}_1\mathbf{e}_2' & E\mathbf{e}_1\mathbf{e}_3' \\ E\mathbf{e}_2\mathbf{e}_1' & E\mathbf{e}_2\mathbf{e}_2' & E\mathbf{e}_2\mathbf{e}_3' \\ E\mathbf{e}_3\mathbf{e}_1' & E\mathbf{e}_3\mathbf{e}_2' & E\mathbf{e}_3\mathbf{e}_3' \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{11}I & \sigma_{12}I & \sigma_{13}I \\ \sigma_{12}I & \sigma_{22}I & \sigma_{23}I \\ \sigma_{13}I & \sigma_{23}I & \sigma_{33}I \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \otimes I_T \\ &= \Sigma \otimes I_T \end{aligned} \qquad (11.2.13)$$

The disturbance covariance matrix $\boldsymbol{\Phi}$ is of dimension $(3T \times 3T)$ with each $(T \times T)$ submatrix being equal to a scalar multiplied by a T-dimensional identity matrix. Kronecker product notation can be used to write the matrix in the convenient form $\Sigma \otimes I_T$. See Section A.15 of Appendix A. An important point to note is that $\boldsymbol{\Phi}$ cannot be written as a scalar multiplied by a $3T$-dimensional identity matrix. Thus, from results in Chapter 8, it follows that the generalized least squares estimator

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (X'\boldsymbol{\Phi}^{-1}X)^{-1}X'\boldsymbol{\Phi}^{-1}\mathbf{y} \\ &= [X'(\Sigma^{-1} \otimes I)X]^{-1}X'(\Sigma^{-1} \otimes I)\mathbf{y} \end{aligned} \qquad (11.2.14)$$

is the best linear unbiased estimator for $\boldsymbol{\beta}$. It has lower variance than the least squares estimator for $\boldsymbol{\beta}$ because it takes into account the contemporaneous correlation between the disturbances in different equations. Because a gain in efficiency can be achieved by combining a number of equations that, at first glance, seem unrelated, Zellner (1962) has given the equations the title of "seemingly unrelated regression equations."

There are two conditions under which least squares is identical to generalized least squares and under which, therefore, there is nothing to be gained by treating the equations as a system. The first is when all contemporaneous correlations are zero. In our example, this condition is

$$\sigma_{12} = \sigma_{13} = \sigma_{23} = 0 \tag{11.2.15}$$

This result is an intuitively reasonable one since it is the existence of these correlations that makes the equations related. The proof that (11.2.15) implies $\mathbf{b} = (X'X)^{-1}X'\mathbf{y} = \hat{\boldsymbol{\beta}} = (X'\Phi^{-1}X)^{-1}X'\Phi^{-1}\mathbf{y}$ is left as a problem (Exercise 11.3.2). The second condition is not so obvious. However, it can be shown that least squares and generalized least squares will yield identical results if the explanatory variables in each equation are identical. In terms of the current example, this condition is

$$X_1 = X_2 = X_3 \tag{11.2.16}$$

Since X_1 contains $\ln p_1$, X_2 contains $\ln p_2$, and X_3 contains $\ln p_3$, this condition clearly does not hold. However, if each equation contained the prices of all three commodities (the cross elasticities were not 0), then (11.2.16) would hold, with each X_i being given by

$$X_i = \begin{bmatrix} 1 & \ln p_{11} & \ln p_{21} & \ln p_{31} & \ln y_1 \\ 1 & \ln p_{12} & \ln p_{22} & \ln p_{32} & \ln y_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \ln p_{1T} & \ln p_{2T} & \ln p_{3T} & \ln y_T \end{bmatrix} \tag{11.2.17}$$

Under these conditions least squares and generalized least squares yield identical results. A proof of this proposition is given in Section 11.2.2.

In Sections 11.2.1 and 11.2.2 we summarize the foregoing results for a general model consisting of M seemingly unrelated regression equations. The question of estimating the unknown covariance matrix Σ is taken up in Section 11.2.3.

11.2.1 General Model Specification

In a general specification of M seemingly unrelated regression equations the ith equation is given by

$$\mathbf{y}_i = X_i\boldsymbol{\beta}_i + \mathbf{e}_i \qquad i = 1, 2, \ldots, M \tag{11.2.18}$$

where $\mathbf{y}_i$ and $\mathbf{e}_i$ are of dimension $(T \times 1)$, X_i is $(T \times K_i)$, and $\boldsymbol{\beta}_i$ is $(K_i \times 1)$. Note that each equation does not have to have the same number of explanatory variables. Combining all equations into one big model yields

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_M \end{bmatrix} = \begin{bmatrix} X_1 & & & \\ & X_2 & & \\ & & \ddots & \\ & & & X_M \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \vdots \\ \boldsymbol{\beta}_M \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \vdots \\ \mathbf{e}_M \end{bmatrix} \tag{11.2.19}$$

or alternatively,

$$\mathbf{y} = X\boldsymbol{\beta} + \mathbf{e} \tag{11.2.20}$$

where the definitions of $\mathbf{y}$, X, $\boldsymbol{\beta}$, and $\mathbf{e}$ are obvious from (11.2.19) and where their dimensions are, respectively, $(MT \times 1)$, $(MT \times K)$, $(K \times 1)$, and $(MT \times 1)$, with $K = \sum_{i=1}^{M} K_i$. Thus the specification (11.2.20) has precisely the form of the linear statistical model considered in earlier chapters.

Given that e_{it} is the error for the ith equation in the tth time period, the assumption of contemporaneous disturbance correlation, but no correlation over time, implies that $E[e_{it}e_{js}] = \sigma_{ij}$ if $t = s$, but 0 if $t \neq s$. Alternatively, $E[\mathbf{e}_i\mathbf{e}_j'] = \sigma_{ij}I_T$, and the covariance matrix for the complete error vector can be written as

$$\Phi = E[\mathbf{e}\mathbf{e}'] = \begin{bmatrix} \sigma_{11}I_T & \sigma_{12}I_T & \cdots & \sigma_{1M}I_T \\ \sigma_{21}I_T & \sigma_{22}I_T & \cdots & \sigma_{2M}I_T \\ \vdots & \vdots & & \vdots \\ \sigma_{M1}I_T & \sigma_{M2}I_T & \cdots & \sigma_{MM}I_T \end{bmatrix} = \Sigma \otimes I_T \tag{11.2.21}$$

where

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1M} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2M} \\ \vdots & \vdots & & \vdots \\ \sigma_{M1} & \sigma_{M2} & \cdots & \sigma_{MM} \end{bmatrix} \tag{11.2.22}$$

The matrix Σ is, of course, symmetric, so that $\sigma_{ij} = \sigma_{ji}$. We will also assume that it is nonsingular and hence positive definite.

In most applications $\mathbf{y}_i$ and X_i, for $i = 1, 2, \ldots, M$, will contain observations on variables for T different time periods, and the subscript i corresponds to a particular economic or geographic unit, such as a household, a firm, or a state. Thus, as mentioned in the introduction to this chapter, the joint model in (11.2.20) can be regarded as one way in which time-series and cross-sectional data can be combined. Other models for combining time-series and cross-sectional data are outlined in Sections 11.4 and 11.5. These models differ depending on the assumptions made about the coefficient and error vectors in each equation.

11.2.2 Estimation with Known Covariance Matrix

When the system (11.2.19) is viewed as the single equation (11.2.20) we can estimate $\boldsymbol{\beta}$ and hence all the $\boldsymbol{\beta}_i$ by the generalized least squares procedures that were discussed in Chapter 8. Thus, using the results of Chapter 8, the generalized least squares estimator

$$\hat{\boldsymbol{\beta}} = (X'\Phi^{-1}X)^{-1}X'\Phi^{-1}\mathbf{y} = [X'(\Sigma^{-1} \otimes I)X]^{-1}X'(\Sigma^{-1} \otimes I)\mathbf{y} \quad (11.2.23)$$

is best linear unbiased. Written in detail, $\hat{\boldsymbol{\beta}}$ becomes

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \\ \vdots \\ \hat{\boldsymbol{\beta}}_M \end{bmatrix} = \begin{bmatrix} \sigma^{11}X_1'X_1 & \sigma^{12}X_1'X_2 & \cdots & \sigma^{1M}X_1'X_M \\ \sigma^{12}X_2'X_1 & \sigma^{22}X_2'X_2 & \cdots & \sigma^{2M}X_2'X_M \\ \vdots & \vdots & & \vdots \\ \sigma^{1M}X_M'X_1 & \sigma^{2M}X_M'X_2 & \cdots & \sigma^{MM}X_M'X_M \end{bmatrix}^{-1} \begin{bmatrix} \sum_{i=1}^{M} \sigma^{1i}X_1'\mathbf{y}_i \\ \sum_{i=1}^{M} \sigma^{2i}X_2'\mathbf{y}_i \\ \vdots \\ \sum_{i=1}^{M} \sigma^{Mi}X_M'\mathbf{y}_i \end{bmatrix} \quad (11.2.24)$$

where σ^{ij} is the (i,j)th element of Σ^{-1}. The covariance matrix for $\hat{\boldsymbol{\beta}}$ is given by $(X'\Phi^{-1}X)^{-1} = [X'(\Sigma^{-1} \otimes I)X]^{-1}$.

If interest centers on one equation, for example, the ith, and only estimators that are a function of $\mathbf{y}_i$ are considered, then the least squares estimator $\mathbf{b}_i = (X_i'X_i)^{-1}X_i'\mathbf{y}_i$ is the minimum variance, linear unbiased estimator. As noted in the preceding paragraph, however, we can improve on this estimator by considering a wider class, namely linear unbiased estimators, that are a function of $\mathbf{y}$. Within this class $\hat{\boldsymbol{\beta}}_i$, the ith vector component of $\hat{\boldsymbol{\beta}}$, is better than $\mathbf{b}_i$ because it allows for the correlation between $\mathbf{e}_i$ and error vectors of the other equations and because it uses information on explanatory variables that are included in the system but excluded from the i th equation. In general, the efficiency gain tends to be higher when the errors among different equations are highly correlated.

There are two cases for the statistical model system (11.2.20) when $\mathbf{b} = \hat{\boldsymbol{\beta}}$ and thus there is no gain in efficiency. One case occurs when Σ is a diagonal matrix, that is, $\sigma_{ij} = 0$ for all $i \neq j$. This situation implies, of course, that there is in fact no correlation between the random errors of different equations.

The second case occurs when the explanatory variables are identical for all equations. That is,

$$X_1 = X_2 = \cdots = X_M = \bar{X} \quad (11.2.25)$$

Under these circumstances $X = I_M \otimes \bar{X}$, and, using rules for the algebra of Kronecker products (Section A.15), we can write

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= [X'(\Sigma^{-1} \otimes I_T)X]^{-1}X'(\Sigma^{-1} \otimes I_T)\mathbf{y} \\ &= [(I_M \otimes \bar{X}')(\Sigma^{-1} \otimes I_T)(I_M \otimes \bar{X})]^{-1}(I_M \otimes \bar{X}')(\Sigma^{-1} \otimes I_T)\mathbf{y} \\ &= [\Sigma^{-1} \otimes \bar{X}'\bar{X}]^{-1}(\Sigma^{-1} \otimes \bar{X}')\mathbf{y} \\ &= [\Sigma \otimes (\bar{X}'\bar{X})^{-1}](\Sigma^{-1} \otimes \bar{X}')\mathbf{y} \\ &= [I_M \otimes (\bar{X}'\bar{X})^{-1}\bar{X}']\mathbf{y} \\ &= [(I_M \otimes \bar{X})'(I_M \otimes \bar{X})]^{-1}(I_M \otimes \bar{X})'\mathbf{y} \\ &= (X'X)^{-1}X'\mathbf{y} = \mathbf{b} \end{aligned} \quad (11.2.26)$$

To verify the step in this proof that uses $[I_M \otimes (\bar{X}'\bar{X})^{-1}\bar{X}'] = (X'X)^{-1}X'$, you should consult the definition of X in (11.2.20) and write out in full the components of $(X'X)^{-1}X'$.

Thus, we have shown that use of the generalized least squares estimator $\hat{\boldsymbol{\beta}}$ does not lead to a gain in efficiency when each equation contains the same explanatory variables. In general, any gain in efficiency tends to be higher when the explanatory variables in the different equations are not highly correlated.

11.2.3 Estimation with Unknown Covariance Matrix

In practice the variances and covariances (σ_{ij}'s) are unknown and must be estimated, with their estimates being used in (11.2.23) to form an estimated generalized least squares estimator. To estimate the σ_{ij} we first estimate each equation by least squares $\mathbf{b}_i = (X_i'X_i)^{-1}X_i'\mathbf{y}_i$ and obtain the least squares residuals $\hat{\mathbf{e}}_i = \mathbf{y}_i - X_i\mathbf{b}_i$. Consistent estimates of the variances and covariances are then given by

$$\hat{\sigma}_{ij} = \frac{1}{T}\hat{\mathbf{e}}_i'\hat{\mathbf{e}}_j = \frac{1}{T}\sum_{t=1}^{T}\hat{e}_{it}\hat{e}_{jt} \quad (11.2.27)$$

Because T is used as the divisor in (11.2.27), the $\hat{\sigma}_{ij}$ will be biased in finite samples. Unlike in the single equation model, there is no completely satisfactory "degrees of freedom correction" that will lead to unbiased variance and covariance estimators for all models. The difficulty is that different equations can have different numbers of regressors and that a divisor that is satisfactory for estimating a variance for one equation is unlikely to be satisfactory for estimating a covariance where two equations are involved. One alternative to using T is to use $T-(K/M)$ where (K/M) is the average number of coefficients per equation. This divisor has the advantage of being constant for the whole system, and it leads to unbiased variance estimates when each equation has the same number of coefficients. The asymptotic properties of the estimated generalized least squares estimator for $\boldsymbol{\beta}$ remain the same irrespective of which (consistent) divisor is used.

If we define $\hat{\Sigma}$ as the matrix Σ with the unknown σ_{ij} replaced by the estimates $\hat{\sigma}_{ij}$, then the corresponding estimated generalized least squares estimator for $\boldsymbol{\beta}$ can be written as

$$\hat{\hat{\boldsymbol{\beta}}} = [X'(\hat{\Sigma}^{-1} \otimes I)X]^{-1}X'(\hat{\Sigma}^{-1} \otimes I)\mathbf{y} \tag{11.2.28}$$

This estimator is the one that is generally used in practice and that is often referred to as Zellner's *seemingly unrelated regression* (*SUR*) *estimator*.

Another estimator for $\boldsymbol{\beta}$ is defined by using (11.2.27) and (11.2.28) in an iterative procedure. A new set of variance estimates can be obtained from

$$\hat{\hat{\sigma}}_{ij} = T^{-1}(\mathbf{y}_i - X_i\hat{\hat{\boldsymbol{\beta}}}_i)'(\mathbf{y}_j - X_j\hat{\hat{\boldsymbol{\beta}}}_j) \tag{11.2.29}$$

where $\hat{\hat{\boldsymbol{\beta}}}' = (\hat{\hat{\boldsymbol{\beta}}}'_1, \hat{\hat{\boldsymbol{\beta}}}'_2, \ldots, \hat{\hat{\boldsymbol{\beta}}}'_M)$. These estimates can be used to form a new estimator for $\boldsymbol{\beta}$, and so on, until convergence. When the random errors follow a multivariate normal distribution this estimator will be the maximum likelihood estimator.

Under appropriate conditions, both estimators, the one based on a single estimate of the disturbance covariance matrix, and the iterative estimator, will have the same limiting distribution. This distribution is approximately normal with mean $\boldsymbol{\beta}$ and a covariance matrix that is consistently estimated by $[X'(\hat{\Sigma}^{-1} \otimes I)X]^{-1}$. Thus, $\hat{\hat{\boldsymbol{\beta}}}$ and the iterative estimator for $\boldsymbol{\beta}$ will be asymptotically more efficient than the least squares estimator. In finite samples, however, there will be a region in the parameter space for Σ where least squares is more efficient than the other estimators. When the contemporaneous correlations are small, the generalized least squares estimator will only be slightly more efficient than the least squares estimator. With the estimated generalized least squares estimator, there is a loss in finite sample efficiency because Σ is replaced by the uncertain estimator $\hat{\Sigma}$. If the contemporaneous correlations are small, this loss in efficiency could be sufficiently great to make least squares better than estimated generalized least squares. The extreme case is when Σ is diagonal. In this case least squares is best. The estimator $\hat{\hat{\boldsymbol{\beta}}}$ can only be as good (in finite samples) if $\hat{\Sigma}$ is diagonal, an extremely unlikely event. If the explanatory variables are identical for each equation, then Σ and $\hat{\Sigma}$ play no role because $\mathbf{b} = \hat{\boldsymbol{\beta}} = \hat{\hat{\boldsymbol{\beta}}}$. When the estimators are numerically equal they are obviously equally efficient.

11.2.4 An Example and Monte Carlo Experiment

As an example, consider the data in Table 11.1, where $\mathbf{x}_{12}$ and $\mathbf{x}_{22}$ are the market values and $\mathbf{x}_{13}$ and $\mathbf{x}_{23}$ are the capital stocks of General Electric and Westinghouse, respectively, for the period 1934 through 1953, as tabulated in Theil (1971). The dependent variables $\mathbf{y}_1$ and $\mathbf{y}_2$ may be regarded as investment figures for the two companies. They are actually artificially generated.

Using least squares estimation for each equation of the system

$$\mathbf{y}_1 = \mathbf{x}_{11}\beta_{11} + \mathbf{x}_{12}\beta_{12} + \mathbf{x}_{13}\beta_{13} + \mathbf{e}_1$$
$$\mathbf{y}_2 = \mathbf{x}_{21}\beta_{21} + \mathbf{x}_{22}\beta_{22} + \mathbf{x}_{23}\beta_{23} + \mathbf{e}_2 \tag{11.2.30}$$

where $\mathbf{x}_{11} = \mathbf{x}_{21}$ are (20×1) vectors of ones, we obtained the following estimates:

$$\mathbf{b}_1 = \begin{bmatrix} 5.279 \\ 0.024 \\ 0.156 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} -0.571 \\ 0.062 \\ 0.072 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} 859.93 & 305.96 \\ 305.96 & 186.15 \end{bmatrix} \tag{11.2.31}$$

Table 11.1 Data for Example Model

$\mathbf{y}_1$	$\mathbf{x}_{12}$	$\mathbf{x}_{13}$	$\mathbf{y}_2$	$\mathbf{x}_{22}$	$\mathbf{x}_{23}$
40.05292	1170.6	97.8	2.52813	191.5	1.8
54.64859	2015.8	104.4	24.91888	516.0	0.8
40.31206	2803.3	118.0	29.34270	729.0	7.4
84.21099	2039.7	156.2	27.61823	560.4	18.1
127.57240	2256.2	172.6	60.35945	519.9	23.5
124.87970	2132.2	186.6	50.61588	628.5	26.5
96.55514	1834.1	220.9	30.70955	537.1	36.2
131.16010	1588.0	287.8	60.69605	561.2	60.8
77.02764	1749.4	319.9	30.00972	617.2	84.4
46.96689	1687.2	321.3	42.50750	626.7	91.2
100.65970	2007.7	319.6	58.61146	737.2	92.4
115.74670	2208.3	346.0	46.96287	760.5	86.0
114.58260	1656.7	456.4	57.87651	581.4	111.1
119.87620	1604.4	543.4	43.22093	662.3	130.6
105.56990	1431.8	618.3	22.87143	583.8	141.8
148.42660	1610.5	647.4	52.94754	635.2	136.7
194.36220	1819.4	671.3	71.23030	723.8	129.7
158.20370	2079.7	726.1	61.72550	864.1	145.5
163.09300	2371.6	800.3	85.13053	1193.5	174.8
227.56340	2759.9	888.9	88.27518	1188.9	213.5

The divisor $T - (K/M)$ was used to obtain $\hat{\Sigma}$. Substituting $\hat{\Sigma}$ into (11.2.28) gives an estimated generalized least squares or seemingly unrelated regression estimate

$$\hat{\hat{\boldsymbol{\beta}}}' = (2.564, 0.024, 0.160, -3.266, 0.065, 0.080)$$

We know that the estimated generalized least squares estimator $\hat{\hat{\boldsymbol{\beta}}}$ is asymptotically superior, or at least not inferior, to the least squares estimator $\mathbf{b}$. However, as we discussed earlier, such is not necessarily the case in small samples. It is interesting, therefore, to investigate the small sample properties of $\mathbf{b}$ and $\hat{\hat{\boldsymbol{\beta}}}$ in a Monte Carlo experiment. Vectors $\mathbf{y}_1$ and $\mathbf{y}_2$, each of size 20 were generated 250 times using the model

$$\begin{aligned} \mathbf{y}_1 &= -28\mathbf{x}_{11} + 0.04\mathbf{x}_{12} + 0.14\mathbf{x}_{13} + \mathbf{e}_1 \\ \mathbf{y}_2 &= -1.3\mathbf{x}_{21} + 0.06\mathbf{x}_{22} + 0.06\mathbf{x}_{23} + \mathbf{e}_2 \end{aligned} \qquad (11.2.32)$$

where the regressors are as in (11.2.30) and the $\mathbf{e}_1$, $\mathbf{e}_2$ are normal random vectors with mean zero and joint variance-covariance matrix

$$E\left[\begin{pmatrix}\mathbf{e}_1 \\ \mathbf{e}_2\end{pmatrix}(\mathbf{e}_1' \quad \mathbf{e}_2')\right] = \boldsymbol{\Phi} = \Sigma \otimes I_{20} = \begin{bmatrix} 660 & 175 \\ 175 & 90 \end{bmatrix} \otimes I_{20} \qquad (11.2.33)$$

Methods for generating multivariate normal random variables are given in the Appendix to this chapter. The first of the 250 generated samples is that given in Table 11.1.

The average values of both the least squares and seemingly unrelated regression estimates together with their standard deviations are presented in Table 11.2. Although we have used samples of only 20 observations, the seemingly unrelated regression estimates have a slightly smaller sampling variance than their least squares counterparts with the exception of β_{13}. This result is also reflected in the frequency distributions in Figure 11.1.

Table 11.2 Means and Standard Deviations of Least Squares and Seemingly Unrelated Estimates from 250 Samples

	Least Squares	SUR
β_{11}	−27.369 (29.505)	−27.455 (28.064)
β_{12}	0.039 (0.0142)	0.039 (0.0135)
β_{13}	0.141 (0.0226)	0.142 (0.0231)
β_{21}	−1.614 (7.8950)	−1.546 (7.3595)
β_{22}	0.060 (0.0155)	0.060 (0.0146)
β_{23}	0.062 (0.0495)	0.063 (0.0489)

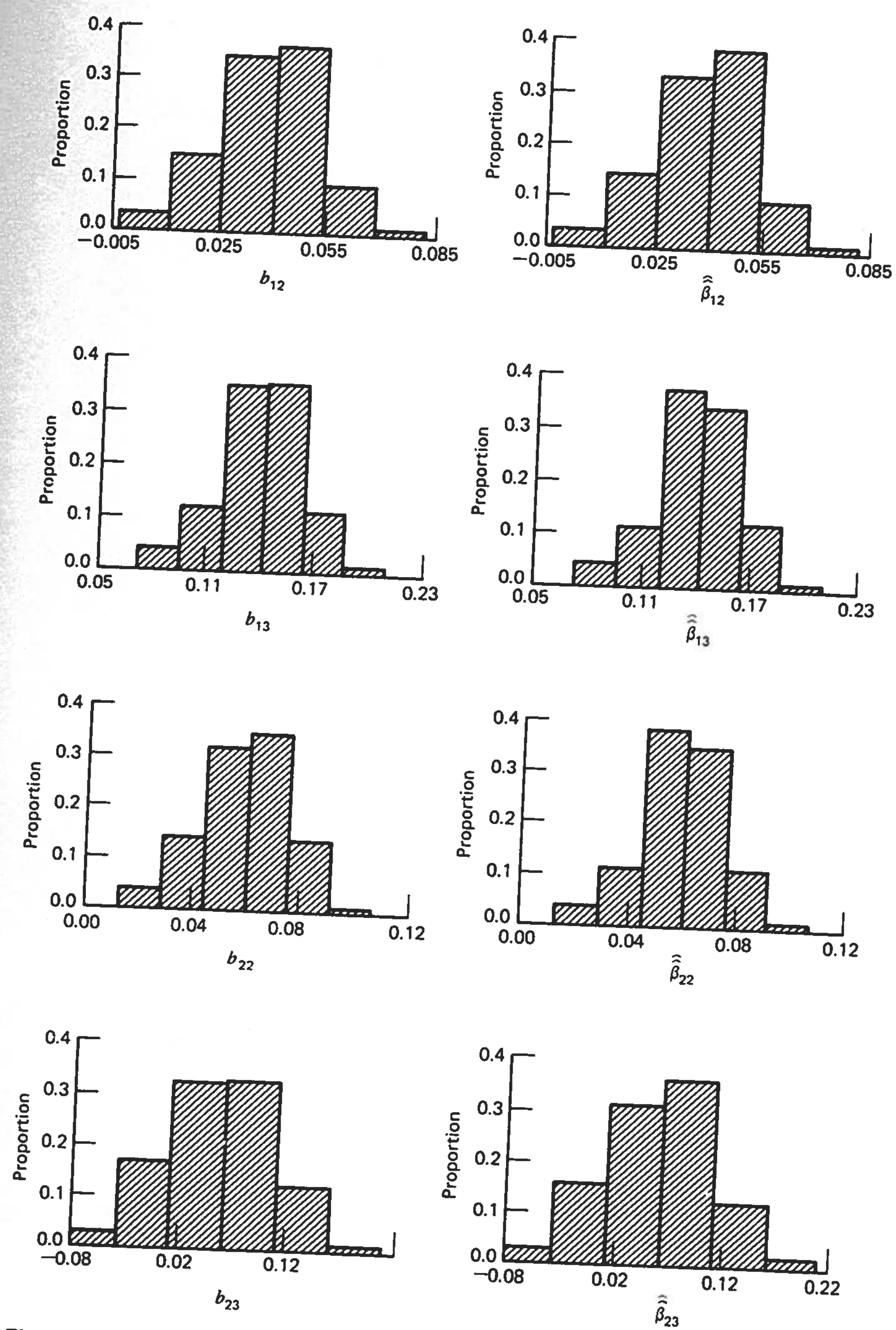

Figure 11.1 Empirical distributions of least squares and seemingly unrelated regression estimates.

11.2.5 Hypothesis Testing and Restricted Estimation

11.2.5a Testing for Contemporaneous Correlation

If contemporaneous correlation does not exist, least squares applied separately to each equation is fully efficient and there is no need to employ the seemingly unrelated regression estimator. Thus, it is useful to test whether the contemporaneous covariances are 0. In the context of the three-equation demand system used to introduce equation systems, the null and alternative hypotheses for this test are

$$H_0: \sigma_{12} = \sigma_{13} = \sigma_{23} = 0$$

$$H_1: \text{at least one covariance is nonzero}$$

An appropriate test statistic is the Lagrange multiplier statistic, suggested by Breusch and Pagan (1980). In the three equation case, this statistic is given by

$$\lambda = T(r_{21}^2 + r_{31}^2 + r_{32}^2) \tag{11.2.34}$$

where r_{ij}^2 is the squared correlation

$$r_{ij}^2 = \frac{\hat{\sigma}_{ij}^2}{\hat{\sigma}_{ii}\hat{\sigma}_{jj}}. \tag{11.2.35}$$

Under H_0, λ has an asymptotic χ^2-distribution with 3 degrees of freedom. Therefore, the null hypothesis is rejected if λ is greater than the critical value from a $\chi^2_{(3)}$-distribution for a prespecified significance level.

In the more general case of M equations, the statistic is given by

$$\lambda = T \sum_{i=2}^{M} \sum_{j=1}^{i-1} r_{ij}^2 \tag{11.2.36}$$

Under H_0, λ has an asymptotic χ^2 distribution with $M(M-1)/2$ degrees of freedom.

11.2.5b Linear Restrictions on the Coefficients

Consider a set of linear restrictions of the form $R\boldsymbol{\beta} = \mathbf{r}$, where R and $\mathbf{r}$ are known matrices of dimensions $(J \times K)$ and $(J \times 1)$, respectively. It is possible to extend the analysis in Sections 6.2 and 6.4 of Chapter 6, and Section 8.5 of Chapter 8, to construct a restricted seemingly unrelated regression estimator and to construct a test statistic for testing the null hypothesis $H_0: R\boldsymbol{\beta} = \mathbf{r}$. There are two main differences between the earlier procedures and those adopted in this section. First, the restricted estimator and the relevant test statistic will now depend on Σ, which, because it is unknown, needs to be replaced by the estimator $\hat{\Sigma}$. This replacement

means that estimator properties and test statistics are based on asymptotic rather than finite sample distributions. Secondly, it is now possible to test and impose restrictions that relate the coefficients in one equation with the coefficients in other equations. This possibility is of particular interest in economics. For example, if the coefficient vectors for each equation are all equal, $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2 = \cdots = \boldsymbol{\beta}_M$, the use of data aggregated over microunits does not lead to aggregation bias (Zellner, 1962). Also, some aspects of economic theory, such as the Slutsky conditions in demand analysis, often suggest symmetric and other linear relationships between coefficients in different equations.

The restricted generalized least squares estimator is obtained by minimizing

$$(\mathbf{y} - X\boldsymbol{\beta})'(\Sigma^{-1} \otimes I)(\mathbf{y} - X\boldsymbol{\beta}) \tag{11.2.37}$$

subject to the linear restrictions $R\boldsymbol{\beta} = \mathbf{r}$. It is given by

$$\hat{\boldsymbol{\beta}}^* = \hat{\boldsymbol{\beta}} + CR'(RCR')^{-1}(\mathbf{r} - R\hat{\boldsymbol{\beta}}) \tag{11.2.38}$$

where

$$C = [X'(\Sigma^{-1} \otimes I)X]^{-1} \tag{11.2.39}$$

and

$$\hat{\boldsymbol{\beta}} = CX'(\Sigma^{-1} \otimes I)\mathbf{y} \tag{11.2.40}$$

When Σ is replaced by the estimator $\hat{\Sigma}$ defined in (11.2.27), we get the restricted seemingly unrelated regression estimator

$$\hat{\hat{\boldsymbol{\beta}}}^* = \hat{\hat{\boldsymbol{\beta}}} + \hat{C}R'(R\hat{C}R')^{-1}(\mathbf{r} - R\hat{\hat{\boldsymbol{\beta}}}) \tag{11.2.41}$$

where

$$\hat{C} = [X'(\hat{\Sigma}^{-1} \otimes I)X]^{-1} \tag{11.2.42}$$

and

$$\hat{\hat{\boldsymbol{\beta}}} = \hat{C}X'(\hat{\Sigma}^{-1} \otimes I)\mathbf{y} \tag{11.2.43}$$

Under the usual assumptions about the limiting behavior of X, and assuming the restrictions $R\boldsymbol{\beta} = \mathbf{r}$ are true, in large samples the distribution of $\hat{\hat{\boldsymbol{\beta}}}^*$ can be approximated by a normal distribution with mean $\boldsymbol{\beta}$ and a covariance matrix that is consistently estimated by $\hat{C} - \hat{C}R'(R\hat{C}R')^{-1}R\hat{C}$. This result is a natural extension of that given in (6.2.17) of Chapter 6.

Turning to the question of testing $H_0: R\boldsymbol{\beta} = \mathbf{r}$ against the alternative $H_1: R\boldsymbol{\beta} \neq \mathbf{r}$, we note that, when H_0 is true,

$$R\hat{\boldsymbol{\beta}} \sim N(\mathbf{r}, RCR') \tag{11.2.44}$$

Thus,

$$g = (R\hat{\boldsymbol{\beta}} - \mathbf{r})'(RCR')^{-1}(R\hat{\boldsymbol{\beta}} - \mathbf{r}) \sim \chi^2_{(J)} \tag{11.2.45}$$

This result is a finite sample one (providing the errors are normally distributed), but it is not operational because it depends on the unknown covariance matrix Σ. When Σ is replaced by $\hat{\Sigma}$, we have

$$\hat{g} = (R\hat{\hat{\boldsymbol\beta}} - \mathbf{r})'(R\hat{C}R')^{-1}(R\hat{\hat{\boldsymbol\beta}} - \mathbf{r}) \xrightarrow{d} \chi^2_{(J)} \tag{11.2.46}$$

Since (11.2.46) only holds when H_0 is true, we reject H_0 if a calculated value for $\hat{g}$ exceeds the appropriate critical value from a $\chi^2_{(J)}$-distribution.

Another way of developing a test statistic for testing $H_0\colon R\boldsymbol\beta = \mathbf{r}$ against $H_1\colon R\boldsymbol\beta \neq \mathbf{r}$ is to begin with an extended version of the single-equation F test that was described in Sections 6.4 and 8.5. Using similar arguments, and assuming the errors are normally distributed, it can be shown that

$$\lambda_F = \frac{g/J}{(\mathbf{y} - X\hat{\boldsymbol\beta})'(\Sigma^{-1}\otimes I)(\mathbf{y} - X\hat{\boldsymbol\beta})/(MT-K)} \sim F_{(J,MT-K)} \tag{11.2.47a}$$

To make this statistic operational, we need to replace Σ by $\hat{\Sigma}$, g by $\hat{g}$, and $\hat{\boldsymbol\beta}$ by $\hat{\hat{\boldsymbol\beta}}$. Then, there are two results that need to be considered. First, the operational statistic that results will only have an approximate $F_{(J,MT-K)}$ distribution. Secondly, it can be shown that the denominator in (11.2.47a) converges in probability to one and hence can be omitted, leaving

$$\hat{\lambda}_F = \frac{\hat{g}}{J} \tag{11.2.47b}$$

as a new operational statistics that has an approximate $F_{(J,MT-K)}$ distribution. Since $J\hat{\lambda}_F = \hat{g} \xrightarrow{d} \chi^2_{(J)}$, and since $JF_{(J,MT-K)} \xrightarrow{d} \chi^2_{(J)}$, the statistics $\hat{g}$ and $\hat{\lambda}_F$ are, in this sense, asymptotically equivalent. In finite samples, use of the F statistic $\hat{\lambda}_F$ rather than the χ^2 statistic $\hat{g}$ will lead to rejection of the null hypothesis in a smaller number of cases. This "more cautious" approach has been recommended by Theil (1971, pp. 402–403); subsequent Monte Carlo studies [see Woodland (1986) and references and discussion therein] have suggested that such caution is justified. Additional caution can be exercised by employing a degrees of freedom correction in the estimation of Σ. If the divisor $T - (K/M)$ is used in place of T, $\hat{g}$ and $\hat{\lambda}_F$ will be smaller and the null hypothesis will be rejected less frequently. For further discussion see Woodland (1986).

Both $\hat{g}$ and $\hat{\lambda}_F$ can be written in terms of restricted and unrestricted residual sums of squares. Such a formulation could be convenient if a computer program that automatically computes $\hat{g}$ or $\hat{\lambda}_F$ is not available. The relationship is (see Exercise 11.3.4)

$$\begin{aligned}\hat{g} &= (R\hat{\hat{\boldsymbol\beta}} - \mathbf{r})'(R\hat{C}R')^{-1}(R\hat{\hat{\boldsymbol\beta}} - \mathbf{r}) \\ &= (\mathbf{y} - X\hat{\hat{\boldsymbol\beta}}^*)'(\hat{\Sigma}^{-1}\otimes I)(\mathbf{y} - X\hat{\hat{\boldsymbol\beta}}^*) - (\mathbf{y} - X\hat{\hat{\boldsymbol\beta}})'(\hat{\Sigma}^{-1}\otimes I)(\mathbf{y} - X\hat{\hat{\boldsymbol\beta}})\end{aligned} \tag{11.2.48}$$

where $\hat{\hat{\boldsymbol\beta}}^*$ is the restricted seemingly unrelated regression estimator defined in (11.2.41). Writing $\hat{g}$ in this way indicates how the test can be viewed as one that examines whether the imposition of restrictions has led to a significant increase in the residual sum of squares.

In addition, rather than use the expression in (11.2.41), it is often easier to obtain the restricted estimator $\hat{\hat{\boldsymbol\beta}}^*$ by rewriting the model (11.2.19) to incorporate the restrictions. For example, if $K_1 = K_2 = \cdots = K_M = k$, and we wish to impose the restrictions $\boldsymbol\beta_1 = \boldsymbol\beta_2 = \cdots = \boldsymbol\beta_M$, (11.2.19) can be written as

$$\begin{bmatrix}\mathbf{y}_1\\ \mathbf{y}_2\\ \vdots\\ \mathbf{y}_M\end{bmatrix} = \begin{bmatrix}X_1\\ X_2\\ \vdots\\ X_M\end{bmatrix}\boldsymbol\beta_1 + \begin{bmatrix}\mathbf{e}_1\\ \mathbf{e}_2\\ \vdots\\ \mathbf{e}_M\end{bmatrix} \tag{11.2.49}$$

or, equivalently, as

$$\mathbf{y} = X^*\boldsymbol\beta_1 + \mathbf{e} \tag{11.2.50}$$

where X^* is of order $(MT \times k)$. The M vector components of $\hat{\hat{\boldsymbol\beta}}^*$ will all be the same, and one of them can be obtained by applying estimated generalized least squares to (11.2.50). This yields

$$\hat{\hat{\boldsymbol\beta}}_1^* = [X^{*\prime}(\hat{\Sigma}^{-1}\otimes I)X^*]^{-1}X^{*\prime}(\hat{\Sigma}^{-1}\otimes I)\mathbf{y} \tag{11.2.51}$$

and the restricted residual sum of squares can be written as

$$(\mathbf{y} - X\hat{\hat{\boldsymbol\beta}}^*)'(\hat{\Sigma}^{-1}\otimes I)(\mathbf{y} - X\hat{\hat{\boldsymbol\beta}}^*) = (\mathbf{y} - X^*\hat{\hat{\boldsymbol\beta}}_1^*)'(\hat{\Sigma}^{-1}\otimes I)(\mathbf{y} - X^*\hat{\hat{\boldsymbol\beta}}_1^*) \tag{11.2.52}$$

A natural question that arises when considering computation of either of the restricted estimator formulations $\hat{\hat{\boldsymbol\beta}}^*$ or $\hat{\hat{\boldsymbol\beta}}_1^*$, is whether Σ should be estimated from unrestricted least squares residuals [as suggested in (11.2.27)] or from restricted least squares residuals. That is, should we use $\hat{\mathbf{e}} = [I - X(X'X)^{-1}X']\mathbf{y}$ or $\mathbf{e}^* = [I - X^*(X^{*\prime}X^*)^{-1}X^{*\prime}]\mathbf{y}$? Since we are interested in the probability distribution of $\hat{g}$ (or of $\hat{\lambda}_F$) when the null hypothesis $R\boldsymbol\beta = \mathbf{r}$ is true, it could be argued that it is more logical to base $\hat{g}$ (or $\hat{\lambda}_F$) on an estimate of Σ that assumes the null hypothesis is true. However, there is no easy way of resolving the question. Only asymptotic results are available and, asymptotically, it makes no difference which estimator for Σ is used. When an iterative procedure is used to estimate $\boldsymbol\beta$ and Σ, the choice between the two estimators for Σ boils down to a choice between the Wald and Lagrange multiplier test statistics. For details of these statistics, and some interesting reformulations, see Judge, et al. (1985, pp. 472–476).

11.2.6 A Further Example

Let us reconsider the three demand equations outlined in (11.2.1), namely,

$$\ln q_{it} = \beta_{i0} + \beta_{ii} \ln p_{it} + \beta_{i4} \ln y_t + e_{it} \qquad i = 1, 2, 3 \qquad (11.2.53)$$

Thirty observations on prices, quantities, and income are given in Table 11.3. These data were used to estimate each equation using least squares, the seemingly unrelated regression (SUR) estimator, and the restricted seemingly unrelated

Table 11.3 Data for Demand Model

p_1	p_2	p_3	y	q_1	q_2	q_3
10.763	4.474	6.629	487.648	11.632	13.194	45.770
13.033	10.836	13.774	364.877	12.029	2.181	13.393
9.244	5.856	4.063	541.037	8.916	5.586	104.819
4.605	14.010	3.868	760.343	33.908	5.231	137.269
13.045	11.417	14.922	421.746	4.561	10.930	15.914
7.706	8.755	14.318	578.214	17.594	11.854	23.667
7.405	7.317	4.794	561.734	18.842	17.045	62.057
7.519	6.360	3.768	301.470	11.637	2.682	52.262
8.764	4.188	8.089	379.636	7.645	13.008	31.916
13.511	1.996	2.708	478.855	7.881	19.623	123.026
4.943	7.268	12.901	433.741	9.614	6.534	26.255
8.360	5.839	11.115	525.702	9.067	9.397	35.540
5.721	5.160	11.220	513.067	14.070	13.188	32.487
7.225	9.145	5.810	408.666	15.474	3.340	45.838
6.617	5.034	5.516	192.061	3.041	4.716	26.867
14.219	5.926	3.707	462.621	14.096	17.141	43.325
6.769	8.187	10.125	312.659	4.118	4.695	24.330
7.769	7.193	2.471	400.848	10.489	7.639	107.017
9.804	13.315	8.976	392.215	6.231	9.089	23.407
11.063	6.874	12.883	377.724	6.458	10.346	18.254
6.535	15.533	4.115	343.552	8.736	3.901	54.895
11.063	4.477	4.962	301.599	5.158	4.350	45.360
4.016	9.231	6.294	294.112	16.618	7.371	25.318
4.759	5.907	8.298	365.032	11.342	6.507	32.852
5.483	7.077	9.638	256.125	2.903	3.770	22.154
7.890	9.942	7.122	184.798	3.138	1.360	20.575
8.460	7.043	4.157	359.084	15.315	6.497	44.205
6.195	4.142	10.040	629.378	22.240	10.963	44.443
6.743	3.369	15.459	306.527	10.012	10.140	13.251
11.977	4.806	6.172	347.488	3.982	8.637	41.845

regression estimator under the restriction that the price elasticities for all commodities are equal ($\beta_{11} = \beta_{22} = \beta_{33}$). Tests for contemporaneous correlation and for the validity of the restrictions $\beta_{11} = \beta_{22} = \beta_{33}$ were also carried out.

Considering first the test for contemporaneous correlation outlined in Section 11.2.5a, we obtain, from the least squares residuals

$$r_{12}^2 = \frac{\hat{\sigma}_{12}^2}{\hat{\sigma}_{11}\hat{\sigma}_{22}} = \frac{(-0.019135)^2}{0.13978 \times 0.18276} = 0.0143$$

$$r_{13}^2 = \frac{\hat{\sigma}_{13}^2}{\hat{\sigma}_{11}\hat{\sigma}_{33}} = \frac{(-0.040329)^2}{0.13978 \times 0.031437} = 0.3701$$

$$r_{23}^2 = \frac{\hat{\sigma}_{23}^2}{\hat{\sigma}_{22}\hat{\sigma}_{33}} = \frac{(-0.037133)^2}{0.18276 \times 0.031437} = 0.2400$$

Then,

$$\lambda = 30(0.0143 + 0.3701 + 0.24) = 18.73$$

The 5% critical value from the χ^2-distribution with 3 degrees of freedom is 7.81. Hence we reject the null hypothesis and conclude that contemporaneous correlation does exist. (From the sample correlations it appears that $\sigma_{23} \neq 0$ and $\sigma_{13} \neq 0$ but that σ_{12} might very well be 0.)

The linear restrictions $\beta_{11} = \beta_{22} = \beta_{33}$ can be written alternatively as $\beta_{11} - \beta_{22} = 0$ and $\beta_{11} - \beta_{33} = 0$. Writing these restrictions in the format $R\boldsymbol{\beta} = \mathbf{r}$ yields

$$\begin{bmatrix} 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} \beta_{10} \\ \beta_{11} \\ \beta_{14} \\ \beta_{20} \\ \beta_{22} \\ \beta_{24} \\ \beta_{30} \\ \beta_{33} \\ \beta_{34} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using the divisor $T - (K/M)$ to estimate Σ, the calculated value of the F-statistic used to test $H_0: R\boldsymbol{\beta} = \mathbf{r}$ against the alternative $H_1: R\boldsymbol{\beta} \neq \mathbf{r}$ (see Equation 11.2.47b) is $\hat{\lambda}_F = 0.569$. The 5% critical value from an F-distribution with (2,81) degrees of freedom is approximately 3.13. Since $0.569 < 3.13$, we accept the null hypothesis.

Four sets of estimates are tabulated in Table 11.4, with the standard errors of the estimates appearing in parentheses below those estimates. Each set of estimates is plausible, with the direct price elasticities all being negative and all income elasticities being positive.

Table 11.4 Demand Equation Estimates

Parameter	Least Squares	SUR	Restricted SUR (Using $\hat{\mathbf{e}}$)	Restricted SUR (Using $\mathbf{e}^*$)
β_{10}	−5.17	−4.57	−4.44	−4.50
	(1.42)	(1.39)	(1.37)	(1.39)
β_{11}	−0.566	−0.909	−0.985	−0.953
	(0.215)	(0.137)	(0.034)	(0.045)
β_{14}	1.434	1.452	1.456	1.455
	(0.229)	(0.228)	(0.228)	(0.230)
β_{20}	−3.64	−3.19	−2.94	−3.00
	(1.61)	(1.58)	(1.56)	(1.58)
β_{22}	−0.648	−0.865	−0.985	−0.953
	(0.188)	(0.132)	(0.034)	(0.046)
β_{24}	1.144	1.137	1.133	1.134
	(0.261)	(0.261)	(0.261)	(0.263)
β_{30}	0.274	0.352	0.321	0.249
	(0.663)	(0.652)	(0.652)	(0.742)
β_{33}	−0.964	−0.999	−0.985	−0.953
	(0.065)	(0.036)	(0.034)	(0.046)
β_{34}	0.871	0.869	0.870	0.871
	(0.108)	(0.108)	(0.108)	(0.123)

Two sets of restricted SUR estimates are presented: those where Σ has been estimated from unrestricted least squares residuals ($\hat{\mathbf{e}}$), and those where Σ has been estimated from restricted least squares residuals ($\mathbf{e}^*$). The divisor $T - (K)$ was used in both cases. The unrestricted SUR estimates are more efficient than least squares, and the restricted SUR estimates are more efficient than the unrestricted SUR estimates. The standard errors give some indication of these efficiency differences, but caution must be exercised because the standard errors are only estimates of the square roots of the true variances.

11.2.7 Sets of Equations with Unequal Numbers of Observations

11.2.7a Theoretical Consequences

So far we have studied a set of equations where the number of observations on each equation was the same. This may not always be the case, however. If we were investigating investment functions for a number of firms, for example, it would not

be surprising to find that the available data on different firms correspond to different time periods. Thus, in this section we allow for the possibility that the available number of observations is different for different equations, and we investigate the implications for the estimation results in Sections 11.2.2 and 11.2.3.

There are two main consequences of the situation outlined here. First, the generalized least squares estimator of the vector containing the coefficients from all the equations can still be obtained in a straightforward manner, but it does not reduce to the same expression obtained when the numbers of observations are equal. Second, the choice of an estimator for the disturbance covariance becomes a problem. Following Schmidt (1977), we illustrate these facts with a system of two seemingly unrelated regression equations

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{bmatrix} = \begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{bmatrix} \tag{11.2.54}$$

where there are T observations on the first equation and $(T + N)$ observations on the second equation. When (11.2.54) is written as

$$\mathbf{y} = X\boldsymbol{\beta} + \mathbf{e} \tag{11.2.55}$$

this implies that $\mathbf{y}$ and $\mathbf{e}$ are of dimension $(2T + N)$, X is $[(2T + N) \times (K_1 + K_2)]$, and $\boldsymbol{\beta}$ is $[(K_1 + K_2) \times 1]$. In line with the earlier assumptions it will be assumed that the vectors $(e_{1t} \;\; e_{2t})'$ are independently and identically distributed with 0 mean and covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \tag{11.2.56}$$

This, in turn, implies that

$$E[\mathbf{e}\mathbf{e}'] = \Phi = \begin{bmatrix} \sigma_{11} I_T & \sigma_{12} I_T & 0 \\ \sigma_{12} I_T & \sigma_{22} I_T & 0 \\ 0 & 0 & \sigma_{22} I_N \end{bmatrix} \neq \Sigma \otimes I_T \tag{11.2.57}$$

and the generalized least squares estimator

$$\hat{\boldsymbol{\beta}} = (X'\Phi^{-1}X)^{-1}X'\Phi^{-1}\mathbf{y} \neq [X'(\Sigma^{-1} \otimes I)X]^{-1}X'(\Sigma^{-1} \otimes I)\mathbf{y} \tag{11.2.58}$$

Thus, although the generalized least squares estimator is readily attainable, it does not reduce to the expression given in (11.2.23). To obtain the corresponding expression we can partition X_2 and $\mathbf{y}_2$ as

$$X_2 = \begin{bmatrix} X_2^* \\ X_2^0 \end{bmatrix} \quad \text{and} \quad \mathbf{y}_2 = \begin{bmatrix} \mathbf{y}_2^* \\ \mathbf{y}_2^0 \end{bmatrix}$$

where $\mathbf{y}_2^*$ contains T observations, $\mathbf{y}_2^0$ contains N observations, and X_2^* and X_2^0 are the corresponding sets of regressors. In this case $\hat{\boldsymbol{\beta}}$ can be written as

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{bmatrix} = \begin{bmatrix} \sigma^{11}X_1'X_1 & \sigma^{12}X_1'X_2^* \\ \sigma^{12}X_2^{*\prime}X_1 & \sigma^{22}X_2^{*\prime}X_2^* + \dfrac{1}{\sigma_{22}}X_2^{0\prime}X_2^0 \end{bmatrix}^{-1} \times \begin{bmatrix} \sigma^{11}X_1'\mathbf{y}_1 + \sigma^{12}X_1'\mathbf{y}_2^* \\ \sigma^{12}X_2^{*\prime}\mathbf{y}_1 + \sigma^{22}X_2^{*\prime}\mathbf{y}_2^* + \dfrac{1}{\sigma_{22}}X_2^{0\prime}\mathbf{y}_2^0 \end{bmatrix} \tag{11.2.59}$$

where σ^{ij} is the (i,j)th element of Σ^{-1}. If the additional N observations $(X_2^0, \mathbf{y}_2^0)$ are ignored, the estimator is identical to (11.2.23).

Because the covariance matrix Σ is usually unknown, we need an estimate for this matrix in order to compute an estimated generalized least squares estimate $\hat{\hat{\boldsymbol{\beta}}}$ of $\boldsymbol{\beta}$. From (11.2.27) the following variance and covariance estimates are suggested.

$$\begin{aligned} \hat{\sigma}_{11} &= \frac{(\mathbf{y}_1 - X_1\mathbf{b}_1)'(\mathbf{y}_1 - X_1\mathbf{b}_1)}{T} \\ \hat{\sigma}_{12} = \hat{\sigma}_{21} &= \frac{(\mathbf{y}_1 - X_1\mathbf{b}_1)'(\mathbf{y}_2^* - X_2^*\mathbf{b}_2)}{T} \\ \hat{\sigma}_{22} &= \frac{(\mathbf{y}_2 - X_2\mathbf{b}_2)'(\mathbf{y}_2 - X_2\mathbf{b}_2)}{T+N} \end{aligned} \tag{11.2.60}$$

where $\mathbf{b}_1$ and $\mathbf{b}_2$ are the least squares estimates of the first and second equations, respectively. Unfortunately, the resulting estimate

$$\hat{\Sigma} = \begin{bmatrix} \hat{\sigma}_{11} & \hat{\sigma}_{12} \\ \hat{\sigma}_{21} & \hat{\sigma}_{22} \end{bmatrix} \tag{11.2.61}$$

may not be positive definite. To avoid this problem, we could ignore the N observations $(X_2^0, \mathbf{y}_2^0)$ and use the estimator $\hat{\Sigma}$ derived in Section 11.2.3. Asymptotically, this estimator is equivalent to (11.2.61). Other possibilities for estimating the covariance matrix Σ are discussed in Judge, et al. (1985, pp. 480–483).

11.2.7b An Example

As an example, we again use the data in Table 11.1 and delete the last five observations of the first equation. Such a situation could arise in practice if the considered companies recorded the data for different time periods. Clearly, in this example $T = 15$ and $N = 5$.

The least squares estimates for the second equation are as in (11.2.31), and

$$\mathbf{b}_1' = (28.463, 0.017, 0.114)$$

The covariance estimate computed acording to (11.2.60) is

$$\hat{\Sigma} = \begin{bmatrix} 756.41 & 282.10 \\ 282.10 & 158.22 \end{bmatrix}$$

Using these variance and covariance estimates in (11.2.59), we get as an estimated generalized least squares estimate,

$$\hat{\hat{\boldsymbol{\beta}}}' = (27.05, 0.013, 0.149, 0.197, 0.057, 0.108)$$

11.2.8 Model Extensions

In Section 11.1 we mentioned that the sets-of-linear-equations model can usually be regarded as one possible model for pooling time-series and cross-sectional data. Given this nature of the data, there are a number of possible extensions that could be made. We saw, in Chapter 9, that the assumption of heteroskedasticity is often a reasonable one when cross-sectional data are being used, and the assumption of autocorrelation is often a reasonable one when time-series data are being used. Because the diagonal elements of the covariance matrix Σ are not necessarily identical, the assumptions of the seemingly unrelated regression model already allow for heteroskedasticity across cross-sectional units. Indeed, the existence of contemporaneous correlation means that allowance is also made for nonzero covariances between the disturbances for different cross-sectional units. However, no allowance for autocorrelation over time has yet been made. A simple way to introduce autocorrelation is to assume that the disturbance in each equation follows the first-order autoregressive process discussed in Section 9.5. One could assume that the autocorrelation coefficient ρ is the same for all equations, or, more realistically, that each equation has a different first-order autoregressive process described by ρ_i, say. When the disturbance in a given equation depends not only on past disturbance values for that equation but also on past values of the disturbances in other equations, the first-order autoregressive process can be generalized to a *vector autoregressive process*. Estimation of these alternative models is discussed in Judge, et al. (1985, pp. 483–493). Some special cases where the coefficient vector is the same for all equations (see Equation 11.2.49) are considered by Kmenta (1986, pp. 618–625). He develops estimators for both Σ diagonal and Σ not diagonal and under the assumption that the errors for each cross-sectional unit follow a first-order autoregressive process.

In a practical setting it is quite likely that extensions such as these will result in a more realistic model. However, it should also be kept in mind that more general models bring with them more parameters to estimate. The greater the number of parameters, the greater the number of observations likely to be needed to obtain reliable estimates. In particular, a reasonable number of time-series observations is needed to estimate an autocorrelation coefficient for each cross-sectional unit. It would be pointless to use one of Kmenta's models if only 10 observations (say) were available on each cross-sectional unit. For further reading on seemingly unrelated regressions, Srivastava and Giles (1987) should be consulted.

11.3 Problems on Seemingly Unrelated Regressions

11.3.1 General Exercises

11.3.1 Prove that application of least squares separately to each of the equations in (11.2.2) yields the same results as application of least squares to the "super model" in (11.2.12).

11.3.2 For the model given in (11.2.11) and (11.2.12), and the definition of Σ given in (11.2.13), prove that $\sigma_{12} = \sigma_{13} = \sigma_{23} = 0$ implies that

$$\mathbf{b} = (X'X)^{-1}X'\mathbf{y} = \hat{\boldsymbol{\beta}} = [X'(\Sigma^{-1} \otimes I)X]^{-1}X'(\Sigma^{-1} \otimes I)\mathbf{y}$$

11.3.3 Show that $\hat{\sigma}_{ij}$ in (11.2.27) is a consistent estimator for σ_{ij} under the assumptions

$$\operatorname{plim} \frac{X_i'\mathbf{e}_j}{T} = \mathbf{0} \qquad \text{for } i, j = 1, 2, \ldots, M$$

and

$$\operatorname{plim} \frac{X_i'X_j}{T} = \lim \frac{X_i'X_j}{T}$$

is finite and nonsingular for $i, j = 1, 2, \ldots, M$.

11.3.4 Prove the result in (11.2.48).

11.3.5 Prove that

$$(\mathbf{y} - X\hat{\hat{\boldsymbol{\beta}}})'(\hat{\Sigma}^{-1} \otimes I)(\mathbf{y} - X\hat{\hat{\boldsymbol{\beta}}}) = \operatorname{tr}[S\hat{\Sigma}^{-1}]$$

where S is an $(M \times M)$ matrix with (i,j)th element equal to $(\mathbf{y}_i - X_i\hat{\hat{\boldsymbol{\beta}}}_i)'(\mathbf{y}_j - X_j\hat{\hat{\boldsymbol{\beta}}}_j)$. If $\hat{\Sigma}$ is based on the residuals of the estimated generalized least squares estimator $\hat{\hat{\boldsymbol{\beta}}}$, rather than the residuals of the least squares estimator $\mathbf{b}$, to what does the foregoing expression simplify?

11.3.6 Consider the following model.

$$\mathbf{y}_1 = \mathbf{x}_1\beta_{11} + \mathbf{x}_2\beta_{12} + \mathbf{e}_1$$
$$\mathbf{y}_2 = \mathbf{x}_3\beta_{21} + \mathbf{x}_4\beta_{22} + \mathbf{e}_2$$

where the covariance matrix for $\mathbf{e} = (\mathbf{e}_1', \mathbf{e}_2')'$ is $\Sigma \otimes I$ with

$$\Sigma = \begin{bmatrix} \sigma^2 & \sigma^2 \\ \sigma^2 & 2\sigma^2 \end{bmatrix}$$

Suppose that data on the dependent and explanatory variables yields

$\mathbf{y}_1'\mathbf{y}_1 = 3000$	$\mathbf{y}_1'\mathbf{y}_2 = 500$	$\mathbf{y}_1'\mathbf{x}_1 = -200$	$\mathbf{y}_1'\mathbf{x}_2 = 400$
$\mathbf{y}_1'\mathbf{x}_3 = 200$	$\mathbf{y}_1'\mathbf{x}_4 = 100$	$\mathbf{y}_2'\mathbf{y}_2 = 1000$	$\mathbf{y}_2'\mathbf{x}_1 = 150$
$\mathbf{y}_2'\mathbf{x}_2 = -200$	$\mathbf{y}_2'\mathbf{x}_3 = 30$	$\mathbf{y}_2'\mathbf{x}_4 = -20$	$\mathbf{x}_1'\mathbf{x}_1 = 100$
$\mathbf{x}_1'\mathbf{x}_2 = 0$	$\mathbf{x}_1'\mathbf{x}_3 = 0$	$\mathbf{x}_1'\mathbf{x}_4 = 0$	$\mathbf{x}_2'\mathbf{x}_2 = 300$
$\mathbf{x}_2'\mathbf{x}_3 = 0$	$\mathbf{x}_2'\mathbf{x}_4 = 0$	$\mathbf{x}_3'\mathbf{x}_3 = 20$	$\mathbf{x}_3'\mathbf{x}_4 = 10$
$\mathbf{x}_4'\mathbf{x}_4 = 10$			

(*a*) Find the best linear unbiased estimates of β_{11}, β_{12}, β_{21}, and β_{22}.
(*b*) Repeat part (*a*) under the restrictions $\beta_{11} = \beta_{21}$ and $\beta_{12} = \beta_{22}$.
(*c*) Test the restrictions imposed in (*b*), given that $T = 15$.

11.3.7 Consider the three-equation demand system

$$\ln q_{it} = \beta_{io} + \beta_{i1} \ln p_{1t} + \beta_{i2} \ln p_{2t} + \beta_{i3} \ln p_{3t} + \beta_{i4} \ln y_t + e_{it}$$
$$i = 1, 2, 3$$

and the data given in Table 11.3. Demand theory suggests that the coefficients satisfy the following constraints:

Engel condition: $w_1\beta_{14} + w_2\beta_{24} + w_3\beta_{34} = 1$

Homogeneity conditions: $\beta_{i1} + \beta_{i2} + \beta_{i3} + \beta_{i4} = 0 \qquad i = 1, 2, 3$

Symmetry conditions: $\dfrac{\beta_{ij}}{w_j} + \beta_{i4} = \dfrac{\beta_{ji}}{w_i} + \beta_{j4} \qquad i, j = 1, 2, 3\ (i \neq j)$

where $w_i = p_iq_i/y$ $(i = 1, 2, 3)$ is the budget share for the ith commodity.
Using the sample means for the budget shares, namely

$$\bar{w}_i = T^{-1} \sum_{t=1}^{T} (p_{it}q_{it}/y_t),$$

(*a*) Write the 7 constraints in the form $R\boldsymbol{\beta} = \mathbf{r}$.
(*b*) Estimate the three equations using both least squares and the restricted seemingly unrelated regression estimator.

(*c*) Test for contemporaneous correlation.
(*d*) Test the validity of the demand theory constraints:
  (*i*) All at once.
  (*ii*) Each group separately.

11.3.2 Exercises Using Monte Carlo Data

Generate 100 samples each of size 20 from the model in (11.2.32) and (11.2.33).

11.3.8 Use five samples of the data to estimate the model using
(*a*) Least squares.
(*b*) Generalized least squares.
(*c*) Estimated generalized least squares.
Compare the various estimates.

11.3.9 Calculate the covariance matrices for the estimators in (*a*) and (*b*) of Exercise 11.3.8.

11.3.10 Using all 100 sets of the estimates obtained in Problem 11.3.8, calculate the mean and variance of each parameter estimate and comment on the results.

11.4 Pooling Time-Series and Cross-Sectional Data Using Dummy Variables

A simple framework that is often used for pooling time-series and cross-sectional data is that provided by dummy variables. In Chapter 10 we saw how dummy variables provide a convenient means of allowing for differences in coefficients that might occur for different samples or for different sample partitions. With pooled time-series and cross-sectional data the different cross-sectional units provide natural sample partitions for which different coefficients or different structures may exist. In this section we are concerned with a model where differences in cross-sectional units can be adequately captured by specifying a different intercept coefficient for each cross-sectional unit. Differences in intercepts are modeled using dummy variables. The model can be viewed as a special case of the seemingly unrelated regression model of Section 11.2, where $\Sigma = \sigma^2 I$ and where the coefficients for all equations are identical, except for the intercepts. Other models are discussed in Sections 11.5 and 11.6.

Assuming we have $i = 1, 2, \ldots, N$ cross-sectional observations, and $t = 1, 2, \ldots, T$ time-series observations, the (i, t)th observation on the dummy variable model with which we are concerned can be written as

$$y_{it} = \beta_{1i} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \tag{11.4.1}$$

where β_{1i} represents the intercept coefficient for the ith cross-sectional unit (or *individual*), the β_k represent the slope coefficients that are common to all individuals, y_{it} is the dependent variable, the x_{kit} are the explanatory variables, and the e_{it} are independent and identically distributed random variables with $E[e_{it}] = 0$ and $E[e_{it}^2] = \sigma_e^2$.

This model is often known as a dummy variable model because it is possible (and convenient) to write it as

$$y_{it} = \sum_{j=1}^{N} \beta_{1j} D_{jt} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \tag{11.4.2}$$

where the D_{jt} are known as dummy variables and take values equal to 0 or 1. Specifically,

$$D_{jt} = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases}$$

Thus there is a dummy variable corresponding to each individual, and the dummy variable that corresponds to individual j will take the value unity for observations on individual j but will be 0 for observations on other individuals.

If we let $\mathbf{j}_T = (1 \quad 1 \cdots 1)'$ be a $(T \times 1)$ vector of ones, then, for the ith individual, (11.4.2) can be written in matrix notation as

$$\mathbf{y}_i = \beta_{1i}\mathbf{j}_T + X_{si}\boldsymbol{\beta}_s + \mathbf{e}_i \qquad i = 1, 2, \ldots, N \tag{11.4.3}$$

where

$$\mathbf{y}_i = \begin{bmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{bmatrix} \qquad X_{si} = \begin{bmatrix} x_{2i1} & x_{3i1} & \cdots & x_{Ki1} \\ x_{2i2} & x_{3i2} & \cdots & x_{Ki2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{2iT} & x_{3iT} & \cdots & x_{KiT} \end{bmatrix} \qquad \mathbf{e}_i = \begin{bmatrix} e_{i1} \\ e_{i2} \\ \vdots \\ e_{iT} \end{bmatrix}$$

and $\boldsymbol{\beta}_s = (\beta_2 \quad \beta_3 \quad \cdots \quad \beta_K)'$. If we let $K' = K - 1$, then it can be noted that $\boldsymbol{\beta}_s$ is a $(K' \times 1)$ vector of slope coefficients and X_{si} is a $(T \times K')$ matrix of observations on the explanatory variables (excluding the constant term) for the ith individual.

The complete set of NT observations can be written as

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_N \end{bmatrix} = \begin{bmatrix} \mathbf{j}_T & \mathbf{0} & \cdots & \mathbf{0} & X_{s1} \\ \mathbf{0} & \mathbf{j}_T & \cdots & \mathbf{0} & X_{s2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{j}_T & X_{sN} \end{bmatrix} \begin{bmatrix} \beta_{11} \\ \beta_{12} \\ \vdots \\ \beta_{1N} \\ \boldsymbol{\beta}_s \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \vdots \\ \mathbf{e}_N \end{bmatrix} \tag{11.4.4}$$

which, using Kronecker product notation, is equivalent to

$$\mathbf{y} = [I_N \otimes \mathbf{j}_T \quad X_s]\begin{pmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_s \end{pmatrix} + \mathbf{e} \tag{11.4.5}$$

where $\mathbf{y}' = (\mathbf{y}_1', \mathbf{y}_2', \ldots, \mathbf{y}_N')$, $X_s' = (X_{s1}', X_{s2}', \ldots, X_{sN}')$, $\mathbf{e}' = (\mathbf{e}_1', \mathbf{e}_2', \ldots, \mathbf{e}_N')$, $\boldsymbol{\beta}_1' = (\beta_{11}, \beta_{12}, \ldots, \beta_{1N})$, and $I_N \otimes \mathbf{j}_T$ is the $(NT \times N)$ matrix of dummy variables

$$I_N \otimes \mathbf{j}_T = \begin{bmatrix} \mathbf{j}_T & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{j}_T & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{j}_T \end{bmatrix} \tag{11.4.6}$$

At this point it is worth emphasizing that, in (11.4.5) the complete $[NT \times (N + K')]$ matrix of explanatory variables

$$[I_N \otimes \mathbf{j}_T \quad X_s]$$

does not contain a constant term.

11.4.1 Parameter Estimation

Of primary interest in (11.4.5) is the estimation of $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_s$. The vector $\boldsymbol{\beta}_1$ contains the N intercept terms, one for each individual, whereas $\boldsymbol{\beta}_s$ is the vector of slope coefficients, which are assumed to be the same for all individuals. Theoretically, this estimation problem does not pose any difficulties. The disturbance vector $\mathbf{e}$ has mean $\mathbf{0}$ and covariance matrix $\sigma_e^2 I_{NT}$, and $[I_N \otimes \mathbf{j}_T \quad X_s]$ is a known nonstochastic matrix of rank $(N + K')$. Therefore, the Gauss-Markov theorem of Chapter 6 is relevant, and the least squares estimator

$$\begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_s \end{bmatrix} = \begin{bmatrix} TI_N & (I_N \otimes \mathbf{j}_T)'X_s \\ X_s'(I_N \otimes \mathbf{j}_T) & X_s'X_s \end{bmatrix}^{-1} \begin{bmatrix} (I_N \otimes \mathbf{j}_T)'\mathbf{y} \\ X_s'\mathbf{y} \end{bmatrix} \tag{11.4.7}$$

is best linear unbiased, with mean $(\boldsymbol{\beta}_1', \boldsymbol{\beta}_s')'$ and covariance matrix

$$\Sigma_{(\mathbf{b}_1', \mathbf{b}_s')} = \sigma_e^2 \begin{bmatrix} TI_N & (I_N \otimes \mathbf{j}_T)'X_s \\ X_s'(I_N \otimes \mathbf{j}_T) & X_s'X_s \end{bmatrix}^{-1} \tag{11.4.8}$$

In both (11.4.7) and (11.4.8) the term TI_N arises because

$$(I_N \otimes \mathbf{j}_T)'(I_N \otimes \mathbf{j}_T) = I_N \otimes \mathbf{j}_T'\mathbf{j}_T = I_N \otimes T = TI_N$$

However, although there are no theoretical problems involved in obtaining $(\mathbf{b}_1' \quad \mathbf{b}_s')$, there could be some numerical problems. The matrix inversion in (11.4.7) is of order $(N + K')$, so if there are many cross-sectional units (N is large), this inversion may be unreliable. Under these circumstances it is advisable to calculate $\mathbf{b}_1$ and $\mathbf{b}_s$ using an alternative expression, derived by taking the *partitioned inverse*, which is discussed in Appendix A.7. The alternative expressions for $\mathbf{b}_1$ and $\mathbf{b}_s$ are (see Exercise 11.7.2)

$$\begin{aligned} \mathbf{b}_s &= (X_s'(I_N \otimes D_T)X_s)^{-1}X_s'(I_N \otimes D_T)\mathbf{y} \\ &= \left(\sum_{i=1}^{N} X_{si}'D_T X_{si}\right)^{-1} \sum_{i=1}^{N} X_{si}'D_T \mathbf{y}_i \end{aligned} \tag{11.4.9}$$

and

$$b_{1i} = \bar{y}_{i.} - \bar{\mathbf{x}}_{i.}'\mathbf{b}_s \qquad i = 1, 2, \ldots, N \tag{11.4.10}$$

where

$$D_T = I_T - \frac{\mathbf{j}_T\mathbf{j}_T'}{T}$$

$$\bar{y}_{i.} = \frac{1}{T}\sum_{t=1}^{T} y_{it} \qquad \bar{\mathbf{x}}_{i.}' = (\bar{x}_{2i.}, \bar{x}_{3i.}, \ldots, \bar{x}_{Ki.}) \tag{11.4.11}$$

and

$$\bar{x}_{ki.} = \frac{1}{T}\sum_{t=1}^{T} x_{kit} \qquad k = 2, 3, \ldots, K$$

In (11.4.9) the matrix inversion is only of order K' and the estimates of the intercepts b_{1i}, $i = 1, 2, \ldots, N$ are straightforward to obtain using (11.4.10).

Let us investigate the nature of the estimator in (11.4.9). It is possible to show that (see Exercise 11.7.1) D_T is idempotent, which implies that $I_N \otimes D_T$ is also idempotent. It is then possible to write

$$\begin{aligned} \mathbf{b}_s &= (X_s'(I_N \otimes D_T)'(I_N \otimes D_T)X_s)^{-1}X_s'(I_N \otimes D_T)'(I_N \otimes D_T)\mathbf{y} \\ &= (Z'Z)^{-1}Z'\mathbf{w} \end{aligned} \tag{11.4.12}$$

where $Z = (I_N \otimes D_T)X_s$ and $\mathbf{w} = (I_N \otimes D_T)\mathbf{y}$ are, respectively, transformed observations on the explanatory variables and transformed observations on the dependent variable. These transformed observations are given by

$$Z = \begin{bmatrix} D_T X_{s1} \\ D_T X_{s2} \\ \vdots \\ D_T X_{sN} \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} D_T \mathbf{y}_1 \\ D_T \mathbf{y}_2 \\ \vdots \\ D_T \mathbf{y}_N \end{bmatrix} \tag{11.4.13}$$

where

$$D_T X_{si} = \begin{bmatrix} x_{2i1} - \bar{x}_{2i.} & \cdots & x_{Ki1} - \bar{x}_{Ki.} \\ x_{2i2} - \bar{x}_{2i.} & \cdots & x_{Ki2} - \bar{x}_{Ki.} \\ \vdots & \ddots & \vdots \\ x_{2iT} - \bar{x}_{2i.} & \cdots & x_{KiT} - \bar{x}_{Ki.} \end{bmatrix} \tag{11.4.14}$$

and

$$D_T \mathbf{y}_i = \begin{bmatrix} y_{i1} - \bar{y}_{i.} \\ y_{i2} - \bar{y}_{i.} \\ \vdots \\ y_{iT} - \bar{y}_{i.} \end{bmatrix} \qquad i = 1, 2, \ldots, N \tag{11.4.15}$$

Thus the matrix D_T, when used to transform the observations on the ith individual (X_{si} and $\mathbf{y}_i$), has the effect of expressing each variable in terms of its deviation from the mean for the ith individual. You are encouraged to verify this fact (Equations 11.4.14 and 11.4.15).

This result implies that, to obtain the least squares estimator of the slope coefficients in the dummy variable model, we can express each variable in terms of deviations from the individual means and run a least squares regression without the constant term. Also, to see that this is an intuitively reasonable procedure, we can average (11.4.1) over time and subtract the result from (11.4.1). This yields

$$y_{it} - \bar{y}_{i.} = \sum_{k=2}^{K} \beta_k (x_{kit} - \bar{x}_{ki.}) + e_{it} - \bar{e}_{i.} \tag{11.4.16}$$

where the definition of $\bar{e}_{i.}$ is obvious, and the estimator $\mathbf{b}_s$ can be viewed as least squares applied to this equation.

To summarize, if we wish to estimate $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_s$ in the dummy variable model in (11.4.5) and if N is small, we can set up the variables as illustrated in (11.4.4) and apply least squares. Alternatively, if N is large, we can proceed in two stages: the slope coefficients $\boldsymbol{\beta}_s$ can be estimated by setting up the variables as shown in (11.4.14) and (11.4.15) and applying least squares; the intercept terms $\boldsymbol{\beta}_1$ can then be estimated by application of (11.4.10).

11.4.2 Variance Estimation

It is worthwhile to give a word of warning concerning estimation of σ_e^2. Both estimation techniques just described lead to the same residual vector. That is, it is possible to show that

$$\hat{\mathbf{e}} = \mathbf{y} - [I_N \otimes \mathbf{j}_T \quad X_s]\begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_s \end{bmatrix} = (I_N \otimes D_T)\mathbf{y} - (I_N \otimes D_T)X_s\mathbf{b}_s \tag{11.4.17}$$

Then, from Chapter 6, an unbiased estimator of σ_e^2 is given by

$$\hat{\sigma}_e^2 = \frac{\hat{\mathbf{e}}'\hat{\mathbf{e}}}{NT - (N + K')} \tag{11.4.18}$$

and this will be the estimate provided by most computer programs if $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_s$ are estimated simultaneously. However, if we use the second estimation technique, where $\mathbf{b}_s$ is obtained as a first step, the variance estimator is likely to be

$$\overset{*}{\sigma}{}_e^2 = \frac{\hat{\mathbf{e}}'\hat{\mathbf{e}}}{NT - K'} \tag{11.4.19}$$

and this will be a biased estimator for σ_e^2. Consequently, under these circumstances it is advisable to "correct" $\overset{*}{\sigma}{}_e^2$ by multiplying by $[(NT - K')/(NT - N - K')]$. The standard errors of the coefficients also need to be adjusted correspondingly.

11.4.3 An Alternative Parameterization

Many computer programs automatically insert a constant term, which often leads novice dummy-variable users to include both a constant term and a dummy variable for each individual. If this happens, part of the matrix of explanatory variables will be of the form

$$[\mathbf{j}_{NT} \quad I_N \otimes \mathbf{j}_T] = \begin{bmatrix} \mathbf{j}_T & \mathbf{j}_T & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{j}_T & \mathbf{0} & \mathbf{j}_T & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{j}_T & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{j}_T \end{bmatrix} \tag{11.4.20}$$

In this matrix the first column (the constant term) is equal to the sum of the next N columns (the dummy variables). Consequently, the columns are linearly dependent or, in other words, the matrix of explanatory variables is not of full rank. This means that the equivalent of the $X'X$ matrix will be singular and that it is impossible to obtain unique estimates of the coefficients.

This problem is a consequence of introducing too many parameters. Before introducing the constant term we already had an intercept for each of the N

individuals; no additional information can be obtained by introducing the constant.

Because of this problem it is often convenient to reparameterize the model so that it includes a constant. We can do this by simply omitting one of the dummy variables. Let us omit the first dummy variable. Then the matrix in (11.4.20) becomes

$$\left[\mathbf{j}_{NT} \quad \begin{pmatrix} \mathbf{0}'_{N-1} \\ I_{N-1} \end{pmatrix} \otimes \mathbf{j}_T \right] = \begin{bmatrix} \mathbf{j}_T & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{j}_T & \mathbf{j}_T & \cdots & \mathbf{0} \\ \vdots & \vdots & & \vdots \\ \mathbf{j}_T & \mathbf{0} & \cdots & \mathbf{j}_T \end{bmatrix} \tag{11.4.21}$$

What effect does this have on our parameter estimates? To investigate this question we rewrite our model (Equation 11.4.2) as

$$\begin{aligned} y_{it} &= \beta_{11}D_{1t} + \sum_{j=2}^{N} \beta_{1j}D_{jt} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \\ &= \delta_1 D_{1t} + \sum_{j=2}^{N} (\delta_1 + \delta_j)D_{jt} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \\ &= \delta_1 \sum_{j=1}^{N} D_{jt} + \sum_{j=2}^{N} \delta_j D_{jt} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \\ &= \delta_1 + \sum_{j=2}^{N} \delta_j D_{jt} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \end{aligned} \tag{11.4.22}$$

The second line in (11.4.22) is obtained by defining $\delta_1 = \beta_{11}$ as the intercept for individual 1 and $\delta_j = \beta_{1j} - \beta_{11}$, $j = 2, 3, \ldots, N$ as the difference between the intercept for the jth individual and that for individual 1. The third line in (11.4.22) uses straightforward algebra, and the last line uses the result that $\sum_{j=1}^{N} D_{jt} = 1$. Also, the final line is in a form that uses (11.4.21) as part of the matrix of explanatory variables. Thus the reparameterized model can be written as

$$\mathbf{y} = \left[\mathbf{j}_{NT} \quad \begin{pmatrix} \mathbf{0}'_{N-1} \\ I_{N-1} \end{pmatrix} \otimes \mathbf{j}_T \quad X_s \right] \begin{bmatrix} \boldsymbol{\delta} \\ \boldsymbol{\beta}_s \end{bmatrix} + \mathbf{e} \tag{11.4.23}$$

where $\boldsymbol{\delta}' = (\delta_1, \delta_2, \ldots, \delta_N)$. Least squares can be applied to this model to obtain estimates $\mathbf{d}$ and $\mathbf{b}_s$, and the only difference between these estimates and those obtained from the previous model is that, with the exception of the first element d_1, $\mathbf{d}$ is a vector of estimates of the differences between the $(N-1)$ intercepts β_{12}, $\beta_{13}, \ldots, \beta_{1N}$ and the intercept for the first individual, β_{11}. If we choose to use (11.4.23) instead of (11.4.5), estimates of the intercepts can be obtained from

$$b_{1j} = d_j + d_1 \qquad j = 2, 3, \ldots, N \tag{11.4.24}$$

and they will be identical to those in the vector $\mathbf{b}_1$ obtained from the original model.

To summarize, it may be more convenient to reformulate the model so that it includes a constant term. This can be achieved by dropping one of the dummy variables, and, when this is done, the new coefficients will have a different interpretation. However, estimates of corresponding coefficients will be identical.

11.4.4 Testing the Dummy Variable Coefficients

A question frequently asked is: Is there evidence to suggest that different individuals have different intercepts, or would the model be adequate if we simply assumed that all the intercepts are identical? If they are all the same and the other assumptions of the model continue to hold, then there is no basis for differentiating the time-series cross-sectional nature of the data, and, for estimation purposes, the data can be treated as one sample of NT observations.

The question can be framed in terms of the hypotheses

$$H_0: \beta_{11} = \beta_{12} = \cdots = \beta_{1N}$$

$$H_1: \text{the } \beta_{1j} \text{ are not all equal}$$

or, using the reparameterized model, in terms of the hypotheses

$$H_0: \delta_2 = \delta_3 = \cdots = \delta_N = 0$$

$$H_1: \text{the } \delta_j \text{ are not all zero}$$

Both null hypotheses represent a set of linear restrictions on coefficients, so they could be formulated in terms of the $R\boldsymbol{\beta} = \mathbf{r}$ framework of Chapter 6. In that chapter we mentioned that when testing a set of linear restrictions, one way to calculate the relevant F statistic is to write it in terms of restricted and unrestricted residual sums of squares. In the context of our hypotheses the F statistic is given by

$$F = \frac{(\bar{\mathbf{e}}'\bar{\mathbf{e}} - \hat{\mathbf{e}}'\hat{\mathbf{e}})/(N-1)}{\hat{\mathbf{e}}'\hat{\mathbf{e}}/(NT - N - K')} \tag{11.4.25}$$

where

1. $\bar{\mathbf{e}}'\bar{\mathbf{e}}$ is the residual sum of squares from the restricted model

$$y_{it} = \beta_{11} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \tag{11.4.26}$$

2. $\hat{\mathbf{e}}'\hat{\mathbf{e}}$ is the residual sum of squares from the unrestricted model

$$y_{it} = \beta_{11} + \sum_{j=2}^{N} \delta_j D_{jt} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \qquad (11.4.27)$$

3. $(N-1)$ is the number of linear restrictions
4. $(NT - N - K')$ is the number of degrees of freedom in the unrestricted model.

Under the null hypothesis the statistic in (11.4.25) has the F distribution with $[(N-1), (NT-N-K')]$ degrees of freedom, and the test is implemented in the usual way.

In applied work it is often tempting to make a judgment about whether some dummy variables should be excluded by examining the t values associated with individual coefficients. Such a practice should not be encouraged. It does provide some information, but, if dummy variables are dropped when t tests are insignificant, two different parameterizations of the same problem can lead to different dummy variables being omitted. For this reason, the joint test in (11.4.25) is preferred.

11.4.5 An Example

Suppose that, for a given industry, we have 10 years ($T = 10$) of data on total cost (y) and output (x) for a sample of four firms ($N = 4$). Also suppose that we wish to estimate the total cost function for each firm and that these functions can be written as

$$y_{it} = \beta_{1i} + \beta_2 x_{it} + e_{it}, \qquad \begin{array}{l} i = 1, 2, 3, 4 \\ t = 1, 2, \ldots, 10 \end{array}$$

where the e_{it} are independent identically distributed random variables with 0 mean and variance σ_e^2. The hypothetical data on these four firms is given in Table 11.5.

When the notation of the previous sections is related to this example, the $(K' \times 1)$ vector $\boldsymbol{\beta}_s$ becomes the scalar, β_2, $\boldsymbol{\beta}_1' = (\beta_{11}, \beta_{12}, \beta_{13}, \beta_{14})$, and the $(NT \times K')$ matrix X_s is simply an $(NT \times 1)$ vector that we will call $\mathbf{x} = (\mathbf{x}_1', \mathbf{x}_2', \mathbf{x}_3', \mathbf{x}_4')'$. Since $N = 4$, this problem is not large, and $\boldsymbol{\beta}_1$ and β_2 could easily be estimated simultaneously by setting up

$$[I_4 \otimes \mathbf{j}_{10} \quad \mathbf{x}]$$

as the matrix of explanatory variables. However, it is instructive to consider the approach based on deviations from firm means, so we will proceed in this direction.

Table 11.5 Data on Cost (y) and Output (x) for Four Firms

Time	Cost				Output			
Period	$i=1$	$i=2$	$i=3$	$i=4$	$i=1$	$i=2$	$i=3$	$i=4$
1	43.72	51.03	43.90	64.29	38.46	32.52	32.86	41.86
2	45.86	27.75	23.77	42.16	35.32	18.71	18.52	28.33
3	4.74	35.72	28.60	61.99	3.78	27.01	22.93	34.21
4	40.58	35.85	27.71	34.26	35.34	18.66	25.02	15.69
5	25.86	43.28	40.38	47.67	20.83	25.58	35.13	29.70
6	36.05	48.52	36.43	45.14	36.72	39.19	27.29	23.03
7	50.94	64.18	19.31	35.31	41.67	47.70	16.99	14.80
8	42.48	38.34	16.55	35.43	30.71	27.01	12.56	21.53
9	25.60	45.39	30.97	54.33	23.70	33.57	26.76	32.86
10	49.81	43.69	46.60	59.23	39.53	27.32	41.42	42.25

For the means we have

$$\bar{y}_{1.} = 36.564 \qquad \bar{y}_{2.} = 43.375 \qquad \bar{y}_{3.} = 31.422 \qquad \bar{y}_{4.} = 47.981$$

$$\bar{x}_{1.} = 30.606 \qquad \bar{x}_{2.} = 29.727 \qquad \bar{x}_{3.} = 25.948 \qquad \bar{x}_{4.} = 28.426$$

and the observations in terms of deviations from the firm means, $(I_4 \otimes D_{10})\mathbf{y}$ and $(I_4 \otimes D_{10})\mathbf{x}$, are given in Table 11.6.

Then, using the estimator in (11.4.9) and (11.4.12), the estimate of the slope coefficient is

$$\begin{aligned} b_2 &= [\mathbf{x}'(I_4 \otimes D_{10})\mathbf{x}]^{-1}\mathbf{x}'(I_4 \otimes D_{10})\mathbf{y} \\ &= (3474.836)^{-1} \cdot 3888.442 \\ &= 1.119029 \end{aligned}$$

and the estimates of the four intercepts are (from Equation 11.4.10)

$$\begin{aligned} b_{11} &= 36.564 - (30.606)(1.119029) = 2.315 \\ b_{12} &= 43.375 - (29.727)(1.119029) = 10.110 \\ b_{13} &= 31.422 - (25.948)(1.119029) = 2.385 \\ b_{14} &= 47.981 - (28.426)(1.119029) = 16.171 \end{aligned}$$

The estimate of the variance of the disturbance term is

$$\hat{\sigma}_e^2 = \frac{\hat{\mathbf{e}}'\hat{\mathbf{e}}}{NT - N - K'} = \frac{491.47896}{35} = 14.042256$$

Table 11.6 Data on Cost and Output in Terms of Deviations from Firm Means

Time Period	Cost				Output			
	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 1$	$i = 2$	$i = 3$	$i = 4$
1	7.156	7.655	12.478	16.309	7.854	2.793	6.912	13.434
2	9.296	−15.625	−7.652	−5.821	4.714	−11.017	−7.428	−0.096
3	−31.824	−7.655	−2.822	14.009	−26.826	−2.717	−3.018	5.784
4	4.016	−7.525	−3.712	−13.721	4.734	−11.067	−0.928	−12.736
5	−10.704	−0.095	8.958	−0.311	−9.776	−4.147	9.182	1.274
6	−0.514	5.145	5.008	−2.841	6.114	9.463	1.342	−5.396
7	14.376	20.805	−12.112	−12.671	11.064	17.973	−8.958	−13.626
8	5.916	−5.035	−14.872	−12.551	0.104	−2.717	−13.388	−6.896
9	−10.964	2.015	−0.452	6.349	−6.906	3.843	0.812	4.434
10	13.246	0.315	15.178	11.249	8.924	−2.407	15.472	13.824

and this leads to a standard error for b_2, given by

$$[\hat{\sigma}_e^2(\mathbf{x}'(I_4 \otimes D_{10})\mathbf{x})^{-1}]^{1/2} = [14.042256 \cdot (3474.836)^{-1}]^{1/2} = 0.06357$$

The four estimated cost functions can now be summarized using one equation with dummy variables, given by

$$y_{it} = 2.315D_{1t} + 10.110D_{2t} + 2.385D_{3t} + 16.171D_{4t} + \underset{(0.064)}{1.119}x_{it} + \hat{e}_{it}$$

We have not provided the standard errors of the coefficients of the dummy variables because, as mentioned in Section 11.4.4, it is usually more meaningful to test them as a group.

With this in mind, we will test the null hypothesis

$$H_0\colon \beta_{11} = \beta_{12} = \beta_{13} = \beta_{14}$$

against the alternative that the β_{1i} are not all equal. To do this we first use the data in Table 11.5 to estimate the restricted model

$$y_{it} = \beta_{11} + \beta_2 x_{it} + e_{it}$$

and obtain

$$y_{it} = 7.385 + 1.132x_{it} + \bar{e}_{it}$$

The residual sum of squares from this estimated equation is

$$\bar{\mathbf{e}}'\bar{\mathbf{e}} = 1838.98022$$

and this leads to the following F statistic (see Equation 11.4.25)

$$F = \frac{(1838.98022 - 491.47896)/3}{491.47896/35} = 31.99$$

For (3, 35) degrees of freedom and a 5 percent significance level, the critical F value is $2.87 < 31.99$, so we reject the null hypothesis and conclude that the intercepts of the four firms' cost functions are not all the same.

11.5 Pooling Time-Series and Cross-Sectional Data Using Error Components

In this section we again consider the model

$$y_{it} = \beta_{1i} + \sum_{k=2}^{K} \beta_k x_{kit} + e_{it} \qquad (11.5.1)$$

but instead of assuming that the β_{1i} are fixed coefficients, we assume that they are independent random variables with a mean $\bar{\beta}_1$ and variance σ_μ^2. We will say more about these alternative assumptions in Section 11.6. For the moment we simply note that, if the β_{1i} are random, it is necessary to view the N individuals as a random sample from some larger population, and it is the population parameters $(\bar{\beta}_1, \beta_2, \ldots, \beta_K)$ that we wish to learn about.

We can write

$$\beta_{1i} = \bar{\beta}_1 + \mu_i \tag{11.5.2}$$

where $E[\mu_i] = 0$, $E[\mu_i^2] = \sigma_\mu^2$, and $E[\mu_i \mu_j] = 0$ for $i \neq j$; and we will also assume that the μ_i are uncorrelated with the e_{jt}. That is, $E[\mu_i e_{jt}] = 0$. Equation 11.5.1 now becomes

$$y_{it} = \bar{\beta}_1 + \sum_{k=2}^{K} \beta_k x_{kit} + \mu_i + e_{it} \tag{11.5.3}$$

which, when written in matrix notation for the ith individual, is

$$\mathbf{y}_i = X_i \boldsymbol{\beta} + \mu_i \mathbf{j}_T + \mathbf{e}_i \tag{11.5.4}$$

where $\mathbf{y}_i$, $\mathbf{j}_T$, and $\mathbf{e}_i$ were defined in the previous section (Equation 11.4.3), $X_i = (\mathbf{j}_T \; X_{si})$ is a $(T \times K)$ matrix of observations on the explanatory variables (including the constant term) for the ith individual, and $\boldsymbol{\beta}' = (\bar{\beta}_1, \beta_2, \ldots, \beta_K)$. In (11.5.4) the term $(\mu_i \mathbf{j}_T + \mathbf{e}_i)$ can be regarded as a composite disturbance vector that has mean $\mathbf{0}$ and covariance matrix

$$\begin{aligned}
V &= E[(\mu_i \mathbf{j}_T + \mathbf{e}_i)(\mu_i \mathbf{j}_T + \mathbf{e}_i)'] \\
&= E[\mu_i^2]\mathbf{j}_T\mathbf{j}_T' + \mathbf{j}_T E[\mu_i \mathbf{e}_i'] + E[\mu_i \mathbf{e}_i]\mathbf{j}_T' + E[\mathbf{e}_i \mathbf{e}_i'] \\
&= \sigma_\mu^2 \mathbf{j}_T \mathbf{j}_T' + \sigma_e^2 I_T \\
&= \begin{bmatrix} \sigma_\mu^2 + \sigma_e^2 & \sigma_\mu^2 & \cdots & \sigma_\mu^2 \\ \sigma_\mu^2 & \sigma_\mu^2 + \sigma_e^2 & \cdots & \sigma_\mu^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_\mu^2 & \sigma_\mu^2 & \cdots & \sigma_\mu^2 + \sigma_e^2 \end{bmatrix}
\end{aligned} \tag{11.5.5}$$

The structure of this covariance matrix is such that, for a given individual, the correlation between any two disturbances in different time periods is the same. Thus in contrast to a first-order autoregressive model, the correlation is constant and does not decline as the disturbances become farther apart in time. Another feature of this matrix is that V does not depend on i, which implies that, not only is the correlation constant over time, it is identical for all individuals.

The complete set of NT observations can be written as

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_N \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix} \boldsymbol{\beta} + \begin{bmatrix} \mu_1 \mathbf{j}_T \\ \mu_2 \mathbf{j}_T \\ \vdots \\ \mu_N \mathbf{j}_T \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \vdots \\ \mathbf{e}_N \end{bmatrix} \tag{11.5.6}$$

or, using obvious definitions, it can be written more compactly as

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\mu} \otimes \mathbf{j}_T + \mathbf{e} \tag{11.5.7}$$

To find the covariance matrix of the composite disturbance in (11.5.7) we note that $E[\boldsymbol{\mu}\boldsymbol{\mu}'] = \sigma_\mu^2 I_N$, $E[\mathbf{e}\mathbf{e}'] = \sigma_e^2 I_{NT}$, and $E[(\boldsymbol{\mu} \otimes \mathbf{j}_T)\mathbf{e}'] = 0$. Then we have

$$\begin{aligned}
\boldsymbol{\Phi} &= E[(\boldsymbol{\mu} \otimes \mathbf{j}_T + \mathbf{e})(\boldsymbol{\mu} \otimes \mathbf{j}_T + \mathbf{e})'] \\
&= \sigma_\mu^2 I_N \otimes \mathbf{j}_T \mathbf{j}'_T + \sigma_e^2 I_{NT} \\
&= I_N \otimes [\sigma_\mu^2 \mathbf{j}_T \mathbf{j}_T' + \sigma_e^2 I_T] \\
&= I_N \otimes V
\end{aligned} \tag{11.5.8}$$

Thus the covariance matrix for the complete $(NT \times 1)$ disturbance vector is block diagonal with each block given by (11.5.5). The block diagonal property arises because the disturbance vectors corresponding to different individuals are uncorrelated. That is,

$$E[(\mu_i \mathbf{j}_T + \mathbf{e}_i)(\mu_j \mathbf{j}_T + \mathbf{e}_j)'] = 0 \qquad \text{for } i \neq j$$

So far in this section we have been describing the assumptions and features of the error components model and, in particular, the structure of its disturbance covariance matrix. We now turn to the problem of estimating $\boldsymbol{\beta}$.

11.5.1 Generalized Least Squares Estimation

It is clear from the foregoing discussion that the covariance matrix for the error components model is not of the scalar-identity type, and, consequently, the model and estimation framework examined in Chapter 8 is relevant. Thus if σ_μ^2 and σ_e^2 (and, consequently, V and $\boldsymbol{\Phi}$) are known, then for estimation of $\boldsymbol{\beta}$ the generalized least squares estimator

$$\hat{\boldsymbol{\beta}} = (X'\boldsymbol{\Phi}^{-1}X)^{-1}X'\boldsymbol{\Phi}^{-1}\mathbf{y} \tag{11.5.9}$$

is best linear unbiased. The easiest way to obtain the generalized least squares estimator is to first find transformed observations $\mathbf{y}^* = P\mathbf{y}$ and $X^* = PX$, where P

is such that $P'P = c\Phi^{-1}$ and c is any scalar. Then, as we have noted in a number of earlier chapters, least squares is applied to the transformed observations to yield

$$\hat{\boldsymbol{\beta}} = (X^{*\prime}X^*)^{-1}X^{*\prime}\mathbf{y}^* = (X'P'PX)^{-1}X'P'P\mathbf{y} = (X'\Phi^{-1}X)^{-1}X'\Phi^{-1}\mathbf{y} \quad (11.5.10)$$

We begin the search for a suitable transformation matrix by noting that

$$\Phi^{-1} = (I_N \otimes V)^{-1} = I_N \otimes V^{-1} \quad (11.5.11)$$

and that (see Exercise 11.7.4)

$$V^{-1} = \frac{\mathbf{j}_T\mathbf{j}_T'}{T\sigma_1^2} + \frac{D_T}{\sigma_e^2} \quad (11.5.12)$$

where $\sigma_1^2 = T\sigma_\mu^2 + \sigma_e^2$ and $D_T = I_T - \mathbf{j}_T\mathbf{j}_T'/T$.
Also, if we define a matrix P_* as

$$P_* = I_T - \alpha\frac{\mathbf{j}_T\mathbf{j}_T'}{T} \quad (11.5.13)$$

where

$$\alpha = 1 - \frac{\sigma_e}{\sigma_1} \quad (11.5.14)$$

then it can be shown that (Exercise 11.7.5) $P_*'P_* = \sigma_e^2V^{-1}$, which implies that the complete transformation matrix is

$$P = I_N \otimes P_* \quad (11.5.15)$$

because

$$\begin{aligned} P'P &= (I_N \otimes P_*)'(I_N \otimes P_*) = I_N \otimes P_*'P_* = I_N \otimes \sigma_e^2\, V^{-1} = \sigma_e^2(I_N \otimes V^{-1}) \\ &= \sigma_e^2\Phi^{-1} \end{aligned}$$

The next question to investigate is the nature of the transformed observations $\mathbf{y}^*$ and X^*. They are given by

$$\mathbf{y}^* = P\mathbf{y} = (I_N \otimes P_*)\mathbf{y} = \begin{bmatrix} P_*\mathbf{y}_1 \\ P_*\mathbf{y}_2 \\ \vdots \\ P_*\mathbf{y}_N \end{bmatrix}$$

and

$$X^* = PX = (I_N \otimes P_*)X = \begin{bmatrix} P_*X_1 \\ P_*X_2 \\ \vdots \\ P_*X_N \end{bmatrix} \quad (11.5.16)$$

Examining the ith vector component of $\mathbf{y}^*$, we have

$$P_*\mathbf{y}_i = \left(I_T - \frac{\alpha\mathbf{j}_T\mathbf{j}_T'}{T}\right)\mathbf{y}_i = \mathbf{y}_i - \alpha\bar{y}_{i.}\mathbf{j}_T \quad (11.5.17)$$

and, similarly, the corresponding component of X^* is

$$P_*X_i = [(1-\alpha)\mathbf{j}_T \quad \mathbf{x}_{2i} - \alpha\bar{x}_{2i.}\mathbf{j}_T \quad \cdots \quad \mathbf{x}_{Ki} - \alpha\bar{x}_{Ki.}\mathbf{j}_T] \quad (11.5.18)$$

where $\mathbf{x}_{ki} = (x_{ki1}, x_{ki2}, \ldots, x_{kiT})'$. Typical elements of (11.5.17) and (11.5.18) are

$$y_{it}^* = y_{it} - \alpha\bar{y}_{i.} \qquad \text{and} \qquad x_{kit}^* = x_{kit} - \alpha\bar{x}_{ki.}$$

Thus the transformed observations are obtained by

1. Calculating the individual means for each variable

$$\bar{y}_{i.} = \frac{\mathbf{j}_T'\mathbf{y}_i}{T} \qquad \text{and} \qquad \bar{x}_{ki.} = \frac{\mathbf{j}_T'\mathbf{x}_{ki}}{T} \qquad k = 2, 3, \ldots, K;\ i = 1, 2, \ldots, N$$

2. Expressing the original observations in terms of deviations from a fraction (α) of their individual means. For the constant term this amounts to replacing a column of ones with a column containing the constant $(1-\alpha)$.

The generalized least squares estimator $\hat{\boldsymbol{\beta}} = (X'\Phi^{-1}X)^{-1}X'\Phi^{-1}\mathbf{y}$ is obtained by applying least squares to the equation

$$y_{it} - \alpha\bar{y}_{i.} = (1-\alpha)\beta_1 + \sum_{k=2}^{K}(x_{kit} - \alpha\bar{x}_{ki.})\beta_k + e_{it} \quad (11.5.19)$$

This can be contrasted with the dummy variable estimator of the slope coefficients, $\mathbf{b}_s$, which is obtained by applying least squares to (see Equation 11.4.16)

$$y_{it} - \bar{y}_{i.} = \sum_{k=2}^{K}(x_{kit} - \bar{x}_{ki.})\beta_k + e_{it} - \bar{e}_{i.} \quad (11.5.20)$$

In fact, if we define $\hat{\boldsymbol{\beta}}_s$ as the generalized least squares estimator of the slope coefficients, obtained by appropriately partitioning $\hat{\boldsymbol{\beta}}$, then it can be shown that $\hat{\boldsymbol{\beta}}_s$ is a matrix weighted average of the dummy variable estimator $\mathbf{b}_s$ and the estimator for $\boldsymbol{\beta}_s$ obtained by carrying out a regression on the individual means. This latter estimator is also encountered in the next section, and we will denote it by $\boldsymbol{\beta}_s^*$. Because it only uses variation between individuals it is often known as the *between estimator*. The generalized least squares estimator can be viewed as an efficient combination of $\boldsymbol{\beta}_s^*$ and the dummy variable estimator $\mathbf{b}_s$, which uses variation within individuals. For more details, see Judge, et al. (1985, p. 523) and references therein.

11.5.2 Estimation of Variance Components

In practice, the generalized least squares estimator $\hat{\boldsymbol{\beta}}$ cannot be used because the variance components σ_μ^2 and σ_e^2 are unknown. Following the practice established in many of the earlier chapters, however, it is possible to replace σ_μ^2 and σ_e^2 with estimates $\hat{\sigma}_\mu^2$ and $\hat{\sigma}_e^2$. When this is done the covariance matrix $\boldsymbol{\Phi}$ is replaced with its estimate $\hat{\boldsymbol{\Phi}}$, and the resulting estimator for $\boldsymbol{\beta}$ is an estimated generalized least squares estimator

$$\hat{\hat{\boldsymbol{\beta}}} = (X'\hat{\boldsymbol{\Phi}}^{-1}X)^{-1}X'\hat{\boldsymbol{\Phi}}^{-1}\mathbf{y} \tag{11.5.21}$$

This estimator is most easily obtained by following the transformation procedure in the previous section, with $\alpha = 1 - \sigma_e/\sigma_1$ replaced by $\hat{\alpha} = 1 - \hat{\sigma}_e/\hat{\sigma}_1$. For conventional estimators of σ_μ^2 and σ_e^2, and under appropriate assumptions about the limiting behavior of X, it can be shown that both $\hat{\boldsymbol{\beta}}$ and $\hat{\hat{\boldsymbol{\beta}}}$ have the same limiting distribution as N or $T \to \infty$.

A large number of alternative estimators for σ_μ^2 and σ_e^2 have been suggested in the literature (Judge, et al., 1985, p. 525); the pair outlined in the following have been chosen because they are unbiased and because they have a certain amount of intuitive appeal.

To estimate σ_e^2 we can use the residuals from the dummy variable estimator. Specifically, an unbiased estimator for σ_e^2 is given by

$$\hat{\sigma}_e^2 = \frac{\hat{\mathbf{e}}'\hat{\mathbf{e}}}{NT - N - K'} \tag{11.5.22}$$

where

$$\hat{\mathbf{e}} = (I_N \otimes D_T)\mathbf{y} - (I_N \otimes D_T)X_s\mathbf{b}_s \tag{11.5.23}$$

The proof that $\hat{\sigma}_e^2$ is unbiased is left as a problem (Exercise 11.7.6).

The next step is to estimate $\sigma_1^2 = T\sigma_\mu^2 + \sigma_e^2$. This is achieved by using the residuals from a regression on the individual means. If we average the original model (Equation 11.5.3) over time we obtain

$$\bar{y}_{i.} = \bar{\beta}_1 + \sum_{k=2}^{K} \beta_k \bar{x}_{ki.} + \mu_i + \bar{e}_{i.} \qquad i = 1, 2, \ldots, N \tag{11.5.24}$$

and the variance of the disturbance term in this equation is

$$\text{var}(\mu_i + \bar{e}_{i.}) = \sigma_\mu^2 + \frac{\sigma_e^2}{T} = \frac{\sigma_1^2}{T} \tag{11.5.25}$$

Also, $(\mu_i + \bar{e}_{i.})$ is uncorrelated with $(\mu_j + \bar{e}_{j.})$ when $i \neq j$. These results imply that the usual least squares estimator obtained from (11.5.24) will be an unbiased estimator of σ_1^2/T. Specifically, (11.5.24) can be written in matrix notation as

$$\bar{\mathbf{y}} = \bar{X}\boldsymbol{\beta} + \mathbf{v} \tag{11.5.26}$$

where $\bar{\mathbf{y}}$ is of dimension $(N \times 1)$ with typical element $\bar{y}_{i.}$, $\bar{X}$ is of dimension $(N \times K)$ with typical element $\bar{x}_{ki.}$, and $\mathbf{v}$ is of dimension $(N \times 1)$ with typical element $\mu_i + \bar{e}_i$. The least squares estimator for $\boldsymbol{\beta}$, from (11.5.26), is

$$\boldsymbol{\beta}^* = (\bar{X}'\bar{X})^{-1}\bar{X}'\mathbf{y} \tag{11.5.27}$$

and the unbiased estimator of σ_1^2/T is

$$\frac{\hat{\sigma}_1^2}{T} = \frac{\mathbf{v}^{*\prime}\mathbf{v}^*}{N - K} \tag{11.5.28}$$

where

$$\mathbf{v}^* = \bar{\mathbf{y}} - \bar{X}\boldsymbol{\beta}^* \tag{11.5.29}$$

An unbiased estimator for σ_μ^2 can then be obtained from

$$\hat{\sigma}_\mu^2 = \frac{\hat{\sigma}_1^2 - \hat{\sigma}_e^2}{T} \tag{11.5.30}$$

A disadvantage of the estimator in (11.5.30) is that it can lead to negative variance estimates. In the event that this occurs, the researcher should rethink the model formulation rather than proceed with the estimated generalized least squares estimator for $\boldsymbol{\beta}$.

In many of the presentations in the literature (e.g., Judge, et al., 1985, p. 523) the between estimator $\boldsymbol{\beta}^*$ is based on NT rather than on N observations because each of the N means is *repeated* T times. This practice will not change the estimator $\boldsymbol{\beta}^*$, but it will make the residual sum of squares T times larger. Consequently, in this case we would be estimating σ_1^2, not σ_1^2/T.

We are now in a position to summarize the estimated generalized least squares procedure.

1. Calculate the dummy variable estimator, either in terms of the dummy variables and original observations or in terms of the observations expressed as deviations from the individual means. The latter estimator is

$$\mathbf{b}_s = [X_s'(I_N \otimes D_T)X_s]^{-1}X_s'(I_N \otimes D_T)\mathbf{y}$$

2. Use the residuals from step 1 to calculate

$$\hat{\sigma}_e^2 = \frac{\hat{\mathbf{e}}'\hat{\mathbf{e}}}{NT - N - K'}$$

3. Estimate $\boldsymbol{\beta}$ using the observations on the individual means

$$\boldsymbol{\beta}^* = (\bar{X}'\bar{X})^{-1}\bar{X}'\bar{\mathbf{y}}$$

4. Use the residuals from step 3 to calculate

$$\frac{\hat{\sigma}_1^2}{T} = \frac{\mathbf{v}^{*\prime}\mathbf{v}^*}{N-K}$$

5. Check to see that $\hat{\sigma}_\mu^2 = (\hat{\sigma}_1^2 - \hat{\sigma}_e^2)/T$ is positive.
6. Calculate $\hat{\alpha} = 1 - \hat{\sigma}_e/\hat{\sigma}_1$.
7. Find the transformed observations

$$y_{it}^* = y_{it} - \hat{\alpha}\bar{y}_{i.} \quad \text{and} \quad x_{kit}^* = x_{kit} - \hat{\alpha}\bar{x}_{ki.}$$

8. Find $\hat{\hat{\boldsymbol{\beta}}}$ by running a least squares regression on the transformed observations.

11.5.3 Prediction of Random Components

In addition to the estimation of $\boldsymbol{\beta}$, we may be interested in predicting the μ_i's, because this prediction describes how the behavior of different individuals varies as well as provides a basis for more efficient prediction of future observations on a given individual. Using the framework of Section 8.7 (see Exercise 11.7.7) it is possible to show that the best linear unbiased predictor for μ_i is

$$\hat{\mu}_i = \left(\frac{\sigma_\mu^2}{\sigma_1^2}\right)\mathbf{j}_T'(\mathbf{y}_i - X_i\hat{\boldsymbol{\beta}}) \tag{11.5.31}$$

and it can be viewed as a proportion of the generalized least squares residual allocation to $\hat{\mu}_i$, the precise proportion depending on the relative variances σ_μ^2 and σ_e^2. It is best linear unbiased in the sense that $E[\hat{\mu}_i - \mu_i] = 0$ and $E[(\hat{\mu}_i - \mu_i)^2]$ is smaller than the prediction error variance of any other predictor that is unbiased and a linear function of $\mathbf{y}$. The expectation is taken with respect to repeated sampling over both time and individuals.

The variance components will, in practice, be unknown, but they can be replaced by the estimates suggested in Section 11.5.2. Replacement yields an intuitively reasonable predictor, but the best linear unbiased property is no longer guaranteed.

11.5.4 Testing the Specification

If $\mu_i = 0$ or, equivalently, $\sigma_\mu^2 = 0$, the individual components do not exist and the least squares estimator is best linear unbiased. To test this hypothesis we could, as mentioned in Section 11.4.4, use the dummy variable estimator and the F test based

on the restricted and unrestricted residual sums of squares. An alternative, which requires only the restricted residuals, is a test based on the Lagrange multiplier statistic (Breusch and Pagan, 1980). Under the null hypothesis $\sigma_\mu^2 = 0$, Breusch and Pagan show that

$$\lambda = \frac{NT}{2(T-1)}\left[\frac{\bar{\mathbf{e}}'(I_N \otimes \mathbf{j}_T\mathbf{j}_T')\bar{\mathbf{e}}}{\bar{\mathbf{e}}'\bar{\mathbf{e}}} - 1\right]^2 \tag{11.5.32}$$

is asymptotically distributed as $\chi^2_{(1)}$, where $\bar{\mathbf{e}}$ is the vector of least squares residuals obtained by regressing $\mathbf{y}$ on X. For computational purposes it is worth noting that

$$\bar{\mathbf{e}}'(I_N \otimes \mathbf{j}_T\mathbf{j}_T')\bar{\mathbf{e}} = \sum_{i=1}^{N}\left(\sum_{t=1}^{T}\bar{e}_{it}\right)^2 \tag{11.5.33}$$

11.5.5 The Example Continued

We now return to the example in Section 11.4.5 where we were estimating the cost functions for four firms. However, instead of assuming that the intercepts are fixed coefficients, we will assume that they are independent random variables with mean $\bar{\beta}_1$ and variance σ_μ^2. Thus we have

$$y_{it} = \bar{\beta}_1 + \beta_2 x_{it} + \mu_i + e_{it}$$

where $i = 1, 2, 3, 4$ and $t = 1, 2, \ldots, 10$. We will first find the estimated generalized least squares estimator for $\boldsymbol{\beta}' = (\bar{\beta}_1\ \beta_2)$ by following the steps summarized at the end of Section 11.5.2.

We have already calculated $\hat{\sigma}_e^2 = 14.042256$. To estimate σ_1^2 we carry out a regression on the four firm means and obtain

$$\bar{y}_{i.} = -2.763 + 1.485\bar{x}_{i.} + v_i^*$$

Then

$$\frac{\hat{\sigma}_1^2}{T} = \frac{\mathbf{v}^{*\prime}\mathbf{v}^*}{N-K} = \frac{133.1507557}{2} = 66.5754$$

and $\hat{\sigma}_1^2 = 665.754$. Checking $\hat{\sigma}_\mu^2$ to see if it is nonnegative, we have

$$\hat{\sigma}_\mu^2 = \frac{\hat{\sigma}_1^2 - \hat{\sigma}_e^2}{10} = 65.17$$

The factor used to transform the observations is

$$\hat{\alpha} = 1 - \frac{\hat{\sigma}_e}{\hat{\sigma}_1} = 0.854768$$

After transforming the observations and applying least squares we have, for the estimated generalized least squares estimates, $\hat{\hat{\beta}}_1 = 7.737534$ and $\hat{\hat{\beta}}_2 = 1.119303$, or

$$\hat{y}_{it} = \underset{(0.063)}{7.738 + 1.119x_{it}}$$

Note that $\hat{\hat{\beta}}_2$ is close to the dummy variable estimator of the slope coefficient $b_2 = 1.119029$. The likely reason for this is the fact that $\hat{\sigma}_1^2$ is large relative to $\hat{\sigma}_e^2$, making $\hat{\alpha}$ close to unity.

Also worth noting is the fact that, in an example such as this in which N is only 4, the estimate of σ_μ^2 is unlikely to be reliable. In fact, in terms of the small sample properties of estimators for $\boldsymbol{\beta}_s$, rather than using an unreliable estimate of σ_μ^2 in an estimated generalized least squares estimator, we are likely to be better off simply using the dummy variable estimator. This, of course, would not be true if the true variances were known.

For predicting the random components μ_1, μ_2, μ_3, and μ_4 we can use the predictor in (11.5.31) with $(\sigma_\mu^2/\sigma_1^2)$ replaced by $(\hat{\sigma}_\mu^2/\hat{\sigma}_1^2)$. This yields

$$\hat{\mu}_1 = \frac{65.17}{665.754}(-54.31) = -5.32$$

$$\hat{\mu}_2 = (0.0979)(23.64) = 2.31$$

$$\hat{\mu}_3 = (0.0979)(-53.59) = -5.25$$

$$\hat{\mu}_4 = (0.0979)(84.26) = 8.25$$

When these answers are compared with the corresponding estimates obtained from the dummy variable estimator they prove to be quite similar. Specifically, we could define a dummy variable estimator of the μ_i as

$$\tilde{\mu}_i = b_{1i} - \frac{1}{4}\sum_{j=1}^{4} b_{1j}, \qquad i = 1, 2, 3, 4$$

and this yields $\tilde{\mu}_1 = -5.43$, $\tilde{\mu}_2 = 2.36$, $\tilde{\mu}_3 = -5.36$, and $\tilde{\mu}_4 = 8.43$.

Finally, let us calculate the value of the Lagrange multiplier test statistic given in (11.5.32). To this end we have $\bar{\mathbf{e}}'(I_N \otimes \mathbf{j}_T\mathbf{j}_T')\bar{\mathbf{e}} = 13470$, $\bar{\mathbf{e}}'\bar{\mathbf{e}} = 1839$, and

$$\lambda = \frac{40}{18}\left[\frac{13470}{1839} - 1\right]^2 = 88.9$$

At the 5 percent significance level the critical value for this test is $\chi^2_{(1)} = 3.84$, so we clearly reject the null hypothesis that $\sigma_\mu^2 = 0$.

11.6 Choice of Model for Pooling

11.6.1 Dummy Variables versus Error Components

Both the dummy variable model (Section 11.4) and the error components model (Section 11.5) can be written as

$$\begin{aligned} \mathbf{y}_i &= X_i\boldsymbol{\beta} + \mu_i\mathbf{j}_T + \mathbf{e}_i \\ &= \mathbf{j}_T\bar{\beta}_1 + X_{si}\boldsymbol{\beta}_s + \mu_i\mathbf{j}_T + \mathbf{e}_i \end{aligned} \tag{11.6.1}$$

In the dummy variable model the μ_i are treated as fixed parameters that need to be estimated, whereas in the error components model the μ_i are treated as a sample of random drawings from a population, and they become part of the model's disturbance term. Least squares (or the "dummy variable estimator") is best linear unbiased for the dummy variable model, and generalized least squares is best linear unbiased for the error components model. We will frame the discussion in terms of the respective estimators for the slope coefficients, $\mathbf{b}_s$ in the case of the dummy variable model and $\hat{\boldsymbol{\beta}}_s$ (or $\hat{\hat{\boldsymbol{\beta}}}_s$) for the error components model.

A number of issues need to be considered when making a choice between fixed or random μ_i (and hence between $\mathbf{b}_s$ and $\hat{\boldsymbol{\beta}}_s$). The first consideration is the relative sizes of N and T. As $T \to \infty$ for fixed N, $\mathbf{b}_s$ and $\hat{\boldsymbol{\beta}}_s$ become identical. Under these circumstances the dummy variable estimator $\mathbf{b}_s$ is consistent and asymptotically efficient, even when the assumptions of the error components model hold. Thus, for large T and small N, there is likely to be little difference between the two estimators, and the natural choice is the one that is computationally easier, namely $\mathbf{b}_s$. When N is large and T is small, the two estimators can differ significantly, and other issues become important. In this case $\mathbf{b}_s$ is still consistent under the assumptions of the error components model, but it is no longer asymptotically efficient, so the question concerning whether the μ_i should be treated as fixed or random needs to be addressed.

One way of viewing the distinction between fixed and random components is as a distinction between conditional and unconditional inference. When the μ_i are fixed, our inference is conditional on the individuals (or cross-sectional units) in the sample. Conditional inference is likely to be appropriate if the individuals on which we have data cannot be regarded as a random sample from some larger population, or if we are particularly interested in those individuals in the sample. If the individuals in the sample can be regarded as a random sample from some larger population, and we are interested in inferences about the population, then unconditional inference that is implicit in the errors components approach is appropriate.

A further consideration is whether the μ_i are likely to be correlated with the X_i. Mundlak (1978) argues that in most applications there is likely to be some relationship between the unmeasurable individual attributes (the μ_i) and the measurable time-varying attributes (the X_i). If such correlation does exist, the estimator $\hat{\boldsymbol{\beta}}_s$ will be biased, but the dummy variable estimator $\mathbf{b}_s$ is not. Thus, conditional inference could be more appropriate if an investigator suspects that μ_i and X_i are correlated. Alternatively, Hausman and Taylor (1981) use knowledge about which vectors in X_i are correlated with the μ_i to develop an efficient instrumental variables estimator. For further information, and an excellent discussion of these issues see Hsiao (1986, pp. 41-52).

As we have mentioned, if N is large and T is small, and the assumptions of the error components model hold, $\hat{\boldsymbol{\beta}}_s$ will be more efficient than $\mathbf{b}_s$. A relevant question is: How large should N be for the estimated generalized least squares estimator $\hat{\hat{\boldsymbol{\beta}}}_s$ to be more efficient than $\mathbf{b}_s$ in finite samples? Taylor (1980) shows that, even in moderately sized samples $[T \geq 3, N-K \geq 9; T \geq 2, N-K \geq 10]$, $\hat{\hat{\boldsymbol{\beta}}}_s$ will be better.

Because both $\mathbf{b}_s$ and $\hat{\boldsymbol{\beta}}_s$ are consistent estimators when the assumptions of the error components model hold, but only $\mathbf{b}_s$ is consistent if, for example, μ_i and X_i are correlated, a test of the appropriateness of the assumptions can be based on whether or not the difference $\mathbf{b}_s - \hat{\boldsymbol{\beta}}_s$ is significant. Hausman (1978) shows that

$$m = (\mathbf{b}_s - \hat{\boldsymbol{\beta}}_s)'(M_1 - M_0)^{-1}(\mathbf{b}_s - \hat{\boldsymbol{\beta}}_s) \qquad (11.6.2)$$

has an asymptotic $\chi^2_{(K')}$ distribution, where $M_1 = \sigma_e^2[X'_s(I_N \otimes D_T)X_s]^{-1}$ is the covariance matrix for the dummy variable estimator $\mathbf{b}_s$ and M_0 is the covariance matrix for the generalized least squares estimator of the slope coefficients $\hat{\boldsymbol{\beta}}_s$. That is, M_0 is the matrix $(X'\Phi^{-1}X)^{-1}$ with its first row and column deleted. The unknown variances in $\hat{\boldsymbol{\beta}}_s$, M_0, and M_1 can be replaced by $\hat{\sigma}_\mu^2$ and $\hat{\sigma}_e^2$ without affecting the asymptotic distribution of m. If the null hypothesis is true, then, asymptotically, $\mathbf{b}_s$ and $\hat{\boldsymbol{\beta}}_s$ differ only through sampling error. However, if μ_i and X_i are correlated, $\hat{\boldsymbol{\beta}}_s$ and $\mathbf{b}_s$ could differ widely, and it is hoped that this will be reflected in the test. Rejection of the null hypothesis suggests that the error components model is not appropriate and that we are likely to be better off using $\mathbf{b}_s$ and regarding the inference as conditional on the μ_i in the sample, or using an estimator that explicitly considers the correlation. Further information on testing is provided by Hausman and Taylor (1981).

11.6.2 Other Models for Pooling Data

The pooling models discussed in this chapter are only a few of the alternatives that are available. Other possibilities make alternative assumptions about the way in which the coefficients change and about the error structure. The seemingly

unrelated regression model can be viewed as a model where all individuals have different coefficient vectors and where these coefficient vectors are regarded as fixed. If the individuals represent a random sample from some larger population, then it might be more appropriate to view the different coefficient vectors as random drawings from a particular probability distribution. This assumption leads to a class of random coefficient models popularized by Swamy (1970).

One possible extension of the dummy variable and error components models is to allow not only for individual-specific effects, but also for time-specific effects. These different effects could be captured by just the intercept term, or they may be applied to all slope coefficients as well. An example of the latter type of model is that studied by Hsiao (1974). Dynamic models, including models with coefficients that evolve over time, have also been suggested in the literature. For a review and summary of the many possible models, refer to Judge, et al. (1985, Chapter 13) and Hsiao (1986).

11.7 Problems on Dummy Variables and Error Components

11.7.1 Algebraic Exercises

11.7.1 Prove that $D_T = I_T - \mathbf{j}_T\mathbf{j}_T'/T$ is an idempotent matrix.

11.7.2 Use results on the partitioned inverse of a matrix to show that (11.4.7) is equivalent to (11.4.9) and (11.4.10).

11.7.3 Prove that the result in (11.4.17) is correct.

11.7.4 Show that the inverse of $V = \sigma_\mu^2\mathbf{j}_T\mathbf{j}_T' + \sigma_e^2 I_T$ is

$$V^{-1} = \frac{\mathbf{j}_T\mathbf{j}_T'}{T\sigma_1^2} + \frac{D_T}{\sigma_e^2}$$

where $\sigma_1^2 = T\sigma_\mu^2 + \sigma_e^2$ and $D_T = I_T - \mathbf{j}_T\mathbf{j}_T'/T$.

11.7.5 Show that $P'_* P_* = \sigma_e^2 V^{-1}$, where V^{-1} is given in Exercise 11.7.4,

$$P_* = I_T - \alpha\mathbf{j}_T\mathbf{j}_T'/T$$

and $\alpha = 1 - \sigma_e/\sigma_1$.

11.7.6 Let $A = I_N \otimes D_T$. Show that (Equation 11.5.23)
(*a*) $\hat{\mathbf{e}} = A\mathbf{y} - AX_s\mathbf{b}_s = (A - AX_s(X_s'AX_s)^{-1}X_s'A)\mathbf{e}$.
(*b*) A has rank $N(T-1)$.

(*c*) $A - AX_s(X_s'AX_s)^{-1}X_s'A$ is idempotent of rank $N(T-1) - K'$.
(*d*) $\hat{\sigma}_e^2$ in (11.5.22) is an unbiased estimator.

11.7.7 In Chapter 8 (Equation 8.7.17) we showed that a general expression for a best linear unbiased predictor is

$$\hat{\mathbf{y}}_0 = X_0\hat{\boldsymbol{\beta}} + V'\boldsymbol{\Psi}^{-1}(\mathbf{y} - X\hat{\boldsymbol{\beta}})$$

For the context of the error components model let $\mathbf{y}_0' = \boldsymbol{\mu}' = (\mu_1, \mu_2, \ldots, \mu_N)$, $X_0 = 0$, $V = E[(\boldsymbol{\mu} \otimes \mathbf{j}_T + \mathbf{e})\boldsymbol{\mu}']$, and $\boldsymbol{\Psi} = \boldsymbol{\Phi}$.
Use this information to show that the best linear unbiased predictor for $\boldsymbol{\mu}$ is given by (11.5.31).

11.7.2 Individual Numerical Exercises

For the remaining problems, generate Monte Carlo data consisting of 50 samples from an error components model with individual effects, namely

$$y_{it} = \bar{\beta}_1 + \beta_2 x_{it} + \mu_i + e_{it}, \qquad \begin{matrix} i = 1, 2, 3, 4 \\ t = 1, 2, \ldots, 10 \end{matrix} \tag{11.7.1}$$

where $\bar{\beta}_1 = 10$ and $\beta_2 = 1$, the μ_i are i.i.d. $N(0, 14)$, and the e_{it} are i.i.d. $N(0, 16)$. The values of the explanatory variable, for the four individuals, are

$$\mathbf{x}_1 = \begin{bmatrix} 38.46 \\ 35.32 \\ 3.78 \\ 35.34 \\ 20.83 \\ 36.72 \\ 41.67 \\ 30.71 \\ 23.70 \\ 39.53 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 32.52 \\ 18.71 \\ 27.01 \\ 18.66 \\ 25.58 \\ 39.19 \\ 47.70 \\ 27.01 \\ 33.57 \\ 27.32 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 32.86 \\ 18.52 \\ 22.93 \\ 25.02 \\ 35.13 \\ 27.29 \\ 16.99 \\ 12.56 \\ 26.75 \\ 41.42 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 41.86 \\ 28.33 \\ 34.21 \\ 15.69 \\ 29.70 \\ 23.03 \\ 14.80 \\ 21.53 \\ 32.86 \\ 42.25 \end{bmatrix}$$

11.7.8 For the model in (11.7.1) calculate the variance of the following estimators for β_2.
(*a*) The least squares estimator $\tilde{b}_2$.
(*b*) The generalized least squares estimator $\hat{\beta}_2$.
(*c*) The dummy variable estimator b_2.
Comment on the relative efficiencies.

11.7.9 Select five samples from the Monte Carlo data and for each sample compute values for the estimators $\hat{\sigma}_1^2$, $\hat{\sigma}_e^2$, and $\hat{\sigma}_\mu^2$. If $\hat{\sigma}_\mu^2 < 0$, set $\hat{\sigma}_\mu^2 = 0$ and $\hat{\sigma}_1^2 = \hat{\sigma}_e^2$. Comment on the values.

11.7.10 For the same five samples compute values for the estimators $\tilde{b}_2$, b_2, and $\hat{\hat{\beta}}_2$, where $\hat{\hat{\beta}}_2$ is the estimated generalized least squares estimator for β_2 that uses the variance estimates obtained in Exercise 11.7.9. Note that b_2 may already have been calculated in Exercise 11.7.9. Compare the estimates with each other and with the true parameter values.

11.7.11 For each of the five samples, test the null hypothesis $\mu_1 = \mu_2 = \mu_3 = \mu_4 = 0$ by using
(*a*) The F test in (11.4.25).
(*b*) The statistic λ in (11.5.32)
In both cases use a 5 percent significance level.

11.7.12 For the same five samples compute estimates for μ_1, μ_2, μ_3, and μ_4 by using
(*a*) The predictor in (11.5.31) with σ_μ^2 and σ_e^2 replaced by $\hat{\sigma}_\mu^2$ and $\hat{\sigma}_e^2$.
(*b*) The dummy variable estimator, that is,

$$\tilde{\mu}_i = b_{1i} - \frac{1}{4}\sum_{j=1}^{4} b_{1j} \qquad i = 1, 2, 3, 4$$

Comment on the values.

11.7.3 Group Exercises

11.7.13 Use all 50 samples to estimate the mean square errors (MSEs) of the estimators $\tilde{b}_2$, b_2, and $\hat{\hat{\beta}}_2$ given in Exercise 11.7.10. With respect to the answers in Exercise 11.7.8, are the MSE estimates of $\tilde{b}_2$ and b_2 accurate? Comment on, and suggest reasons for, the relative efficiencies of $\tilde{b}_2$, b_2, and $\hat{\hat{\beta}}_2$.

11.7.14 Repeat Exercise 11.7.11 for all 50 samples. Which test appears to be more powerful?

11.8 References

Breusch, T. S., and A. R. Pagan (1980) "The LaGrange Multiplier Test and Its Applications to Model Specification in Econometrics," *Review of Economic Studies*, 47, 239–254.

Hausman, J. A. (1978) "Specification Tests in Econometrics." *Econometrica*, 46, 1251–1272.

Hausman, J. A., and W. E. Taylor (1981) "Panel Data and Unobservable Individual Effects." *Econometrica*, 49, 1377–1398.

Hsiao, C. (1974) "Statistical Inference for a Model with Both Random Cross Sectional and Time Effects." *International Economic Review*, 15, 12–30.

Hsiao, C. (1986) *Analysis of Panel Data.* Cambridge, Eng: Cambridge University Press.

Judge, G. G., W. E. Griffiths, R. C. Hill, H. Lütkepohl, and T. C. Lee (1985) *The Theory and Practice of Econometrics*, 2nd ed., New York: Wiley.

Kmenta, J. (1986) *Elements of Econometrics*, 2nd ed., New York: Macmillan.

Mundlak, Y. (1978) "On the Pooling of Time Series and Cross Section Data." *Econometrica*, 46, 69–85.

Schmidt, P. (1977) "Estimation of Seemingly Unrelated Regressions with Unequal Numbers of Observations." *Journal of Econometrics*, 5, 365–377.

Srivastava, V. K. and D. E. A. Giles (1987) *Seemingly Unrelated Regression Models: Estimation and Inference.* New York: Dekker.

Swamy, P. A. V. B. (1970) "Efficient Inference in a Random Coefficient Regression Model." *Econometrica*, 38, 311–323.

Taylor, W. E. (1980) "Small Sample Considerations in Estimation from Panel Data." *Journal of Econometrics*, 13, 203–223.

Theil, H. (1971) *Principles of Econometrics.* New York: Wiley.

Woodland, A. D. (1986) "An Aspect of the Wald Test for Linear Restrictions in the Seemingly Unrelated Regressions Model." *Economics Letters*, 20, 165–169.

Zellner, A. (1962) "An Efficient Method of Estimating Seemingly Unrelated Regressions and Tests of Aggregation Bias." *Journal of the American Statistical Association*, 57, 348–368.

11.A Appendix: Bivariate (Multivariate) Normal Random Variables

Bivariate normal random variables may be generated by the following procedures: Let Σ be the covariance matrix, C be the corresponding matrix of characteristic vectors, and D be the diagonal matrix of characteristic roots. Then $C\Sigma C' = D$ and $D^{-1/2}C\Sigma C'D^{-1/2} = D^{-1/2}DD^{-1/2} = I$ and $\Sigma = C'D^{1/2}D^{1/2}C$.

As an example, assume that the covariance matrix is

$$\Sigma = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

with the corresponding matrix of characteristic vectors

$$C = \begin{bmatrix} 0.5257 & 0.8506 \\ 0.8506 & -0.5257 \end{bmatrix}$$

and the diagonal matrix of characteristic roots 2.6180 and 0.3820. Let

$$D^{1/2} = \begin{bmatrix} \sqrt{2.6180} & 0 \\ 0 & \sqrt{0.3820} \end{bmatrix} = \begin{bmatrix} 1.6180 & 0 \\ 0 & 0.6180 \end{bmatrix}$$

Thus

$$C'D^{1/2} = \begin{bmatrix} 0.8506 & 0.5257 \\ 1.3763 & -0.3249 \end{bmatrix}$$

Given the foregoing matrix, each pair of generated independent standard normal random variates, e_1 and e_2 are used with $C'D^{1/2}$ to obtain bivariate normal variates e_1^* and e_2^*.

$$\begin{bmatrix} 0.8506 & 0.5257 \\ 1.3763 & -0.3249 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} e_1^* \\ e_2^* \end{bmatrix}$$

The bivariate normal random variates e_1^* and e_2^* have mean

$$E\begin{bmatrix} e_1^* \\ e_2^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and covariance matrix

$$E\begin{bmatrix} e_1^* \\ e_2^* \end{bmatrix}[e_1^* \quad e_2^*] = C'D^{1/2}\left(E\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}[e_1 \quad e_2]\right)D^{1/2}C$$

$$= \begin{bmatrix} 0.8506 & 0.5257 \\ 1.3763 & -0.3249 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.8506 & 1.3763 \\ 0.5257 & -0.3249 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \Sigma$$

Alternatively, if a computer package such as SHAZAM is available, the Cholesky matrix decomposition method may be used to obtain a nonsingular triangular matrix P such that $PP' = \Sigma$ or

$$P\left(E\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}[e_1 \quad e_2]\right)P' = \Sigma$$

Consequently,

$$P\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} e_1^* \\ e_2^* \end{bmatrix}$$

For example, if

$$\Sigma = \begin{bmatrix} 3 & 2 \\ 2 & 8 \end{bmatrix}$$

this yields a Cholesky matrix

$$P = \begin{bmatrix} 1.732051 & 0 \\ 1.154701 & 2.581989 \end{bmatrix}$$

and

$$P\left(E\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}[e_1 \quad e_2]\right)P' = \begin{bmatrix} 1.732051 & 0 \\ 1.154701 & 2.581989 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1.732051 & 1.154701 \\ 0 & 2.581989 \end{bmatrix}$$
$$= \begin{bmatrix} 3 & 2 \\ 2 & 8 \end{bmatrix} = \Sigma$$

In a Monte Carlo sampling study, using the transformation matrix, these procedures are repeated for each of T pairs of random variables for each of the N samples.

CHAPTER 12

Nonlinear Least Squares and Nonlinear Maximum Likelihood Estimation

12.1 Introduction

Nonlinearities enter economic models in various forms. As we have seen in Chapters 5 and 6, if only the variables enter nonlinearly, the model can still be handled in the linear model framework. Furthermore, if the nonlinearity is in the parameters or in the variables and the parameters, it is sometimes possible to find a transformation to convert the considered model into a linear specification. Generally, however, this is not possible, and therefore we discuss a model of the general form

$$y_t = f(\mathbf{x}_t, \boldsymbol{\beta}) + e_t \tag{12.1.1}$$

in this chapter. In (12.1.1) $\mathbf{x}_t$ is an $(N \times 1)$ vector of independent variables, $\boldsymbol{\beta}$ is a $(K \times 1)$ parameter vector, y_t is the dependent variable whose mean is a function of $\mathbf{x}_t$ and $\boldsymbol{\beta}$, and e_t is a random error.

One example is the Cobb–Douglas production function

$$Q_t = \alpha L_t^{\beta_1} K_t^{\beta_2} + e_t \tag{12.1.2}$$

where Q_t is the output from a section of the economy, L_t is the labor input, and K_t is the capital input, all in period t. The parameters are α, β_1, and β_2. The latter two are the elasticities of Q_t with respect to L_t and K_t. Defining $y_t = Q_t$, $\mathbf{x}_t' = (L_t\ K_t)$, $\boldsymbol{\beta}' = (\alpha\ \beta_1\ \beta_2)$, and $f(\mathbf{x}_t, \boldsymbol{\beta}) = \alpha L_1^{\beta_1} K_t^{\beta_2}$, the Cobb–Douglas production function (12.1.2) has exactly the form (12.1.1). Note that, unlike linear specifications, the number of parameters, K, and the number of independent variables, N, do not necessarily coincide in nonlinear models.

Another example is a consumption function

$$C_t = \beta_1 + \beta_2 Y_t^{\beta_3} + e_t \tag{12.1.3}$$

which could arise if there is uncertainty as to the way the consumption expenditures C_t depend on the income Y_t. Why should the form

$$C_t = \beta_1 + \beta_2 Y_t + e_t$$

be a priori more plausible than

$$C_t = \beta_1 + \beta_2 Y_t^{1/2} + e_t$$